Artificial Intelligence Review (2025) 58:368
https://doi.org/10.1007/s10462-025-11357-w

Effectiveness of data resampling and ensemble learning in multiclass imbalance learning

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Accepted: 14 August 2025 / Published online: 6 October 2025

Abstract

Classification tasks in many real-world problems often involve multiclass datasets with imbalanced class distributions, which have more difficulty factors than binary classification. Previous studies have proposed various methods to address this multiclass imbalanced learning issue. Data resampling and ensemble learning are the most popular among the proposed methods. However, no comprehensive review or survey has provided an in-depth comparison of ad hoc methods in multiclass imbalance learning, particularly with a focus on data resampling and ensemble learning. Moreover, there is a lack of studies that analyze the effectiveness of each method in terms of the difficulty factors in multiclass imbalance learning. This paper provides a comprehensive review and comparative analysis to identify the strengths and weaknesses of each method and assess their effectiveness in improving classification performance. The analysis shows that not all methods effectively enhance classification performance on multiclass imbalanced datasets. Some methods even perform worse than the baseline performance. The review also reveals that datasets with certain difficulty factors are more challenging for most existing methods to handle. Ultimately, this paper summarizes several important lessons and identifies research gaps to guide future work in the field.

Keywords Multiclass imbalance · Imbalance learning · Data resampling · Ensemble learning

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1 Introduction

Imbalance learning is one of the most common challenges in developing classification models using machine learning algorithms, as they often work effectively only on balanced datasets (Sun et al. 2009). Numerous classification cases face the issue of class imbalance, such as data classification in the fields of healthcare (Sujay et al. 2024; Cusworth et al. 2024), network security (Zoghi and Serpen 2024; Zhou et al. 2025), finance (Khalil et al. 2024; Chhabra et al. 2024), manufacturing (Hirsch et al. 2020; Dasari et al. 2022), telecommunication (Li et al. 2023; Aravinda et al. 2024), and business (Wong et al. 2020; Teng et al. 2024).

Most of the previous studies focused on imbalance learning on binary class datasets. However, real-world problems also involve datasets with multiple classes (Sridhar and Kalaivani 2021). Although methods for handling imbalanced binary-class datasets can be adapted for multiclass imbalance problems, such as using the one-versus-all (OvA) or one-versus-one (OvO) approaches, they are less effective (Lango and Stefanowski 2022). This is due to the characteristic differences between the two types of datasets, where multiclass datasets have higher data complexity than binary class ones. Therefore, the imbalance problem in multiclass datasets requires ad hoc methods that can specifically address existing problems.

The main problem with imbalanced datasets is not only the lack of data on minority classes but also the presence of some difficulty factors (Vuttipittayamongkol et al. 2021). To date, no study on the multiclass imbalance problem has evaluated the performance of ad hoc methods in relation to these difficulty factors. According to He and Garcia (2009), class overlap and small disjuncts are the most significant factors that deteriorate classification performance in binary class imbalance learning. In the case of multiclass imbalance learning, several additional difficulty factors increase the complexity of the problem, namely the class configuration and the number of classes in the dataset (Lango and Stefanowski 2022). The class configuration corresponds to the composition of majority and minority classes within the dataset. The number of classes within the dataset also influences classification performance in multiclass imbalance learning. As Lango and Stefanowski (2022) noted, increasing the number of classes tends to decrease the recall of instances from minority classes.

Research on multiclass imbalance learning has increased significantly over the last five years compared to the previous period. Most of them are dominated by works on data resampling and ensemble learning. Several ad hoc resampling methods have been proposed, employing different approaches, such as interpolation-based resampling, boundary-based resampling, and other complex mechanisms that integrate data clustering and neighborhood selection to ensure that synthetic instances are generated in the appropriate area. On the other hand, several ensemble learning approaches based on bagging and boosting mechanisms have been proposed. To enhance its performance, most ensemble methods integrate the data resampling technique within the ensemble mechanism to balance the class distribution. SMOTE is a common method used in ensemble approaches.

The various strategies and mechanisms proposed for addressing the multiclass imbalance learning problem require thorough evaluation and review to identify unresolved challenges and gaps in the current research. Given these challenges, this review paper is necessary to bridge these gaps and serve as a valuable reference for future studies and developments in the field. Many review and survey articles have been published as the field of research continues to grow rapidly. However, critical research gaps remain, requiring further investigation, including:

1. A lack of comprehensive literature examining ad hoc methods in multiclass imbalance learning in depth.
2. An absence of studies that assess and analyze the extent of performance improvements achieved by existing ad hoc methods compared to baseline performance.
3. A lack of literature evaluating the relationship between method performance and difficulty factors in multiclass imbalance learning.

Therefore, this paper compiles and analyzes existing ad hoc methods, including data resampling and ensemble learning approaches, to derive valuable insights and contribute to the field's body of knowledge. The main contributions of this paper are as follows:

1. Providing a comprehensive review of ad hoc methods, focusing on data resampling and ensemble learning, two of the most widely used approaches. Most of the methods reviewed in this study have not been discussed in depth in previous studies. The review includes a critical analysis and empirical evaluation of the performance of ad hoc methods. This helps in understanding the advantages and limitations of each method.
2. Providing an effectiveness assessment by conducting a performance analysis against the baseline performance to evaluate the improvements achieved by each method. The analysis is conducted through performance comparison among ad hoc methods based on relevant metrics used in imbalanced classification. This can help evaluate the performance of each method and identify areas for future improvement.
3. Providing relationship analysis between method performance and difficulty factors. The analysis includes the performance of ad hoc methods under various imbalance ratios, class overlap percentages, class numbers, and class configurations. Understanding this relationship enables the identification of optimal strategies to address multiclass imbalanced datasets using data resampling or ensemble learning.
4. Providing insights and research gaps for future works on data resampling and ensemble learning in multiclass imbalance learning. Future work includes the development of oversampling and undersampling mechanisms, feature engineering, voting mechanisms, and the application of specific base classifiers for ensemble learning.

The rest of this paper is arranged as follows. Section 2 discusses previous review and survey papers on multiclass imbalance learning. Section 3 provides a general overview of imbalanced classification to support understanding its core problems and advancements. Section 4 discusses the key challenges in multiclass imbalance learning, including multiclass overlap, small disjuncts, class configuration, and the number of classes. Section 5 provides an overview of the state-of-the-art methods in multiclass imbalance learning, covering data resampling and ensemble learning approaches in detail. This section further breaks down resampling methods based on resampling scope, step, and mechanisms. It further explores ensemble learning methods based on mechanisms, algorithms, base classifiers, data processing strategies, and aggregation methods. Section 6 reviews the datasets commonly used in research on multiclass imbalance learning. Section 7 explains various performance metrics used in recent studies. Section 8 evaluates the effectiveness of these state-of-the-art methods, with subsections that compare the performance of the resampling and ensemble learning approaches against each other and the baseline performance. Section 9 summarizes the lessons learned from the review and analysis and proposes directions for future research. Finally, Sect. 10 concludes the paper.

2 Related work

In this work, we collect 24 review and survey papers on imbalance learning from reputable Scopus-indexed journals published between 2017 and 2024. This initial search encompasses topics related to binary and multiclass imbalance learning. To obtain credible and high-quality information, we limit the search to Scopus-indexed journals in the first and second quartiles and exclude conference articles. Those articles come from seven publishers: Springer (10 articles), Elsevier (7 articles), ACM (2 articles), IEEE (2 articles), MDPI (1 article), Wiley (1 article), and OUP (1 article).

Most (18 out of 24) of the collected articles are on the binary class imbalance problem. In general, these articles only discuss the trend and application of imbalance learning in various fields (Kaur et al. 2019; Lin et al. 2021; Chen et al. 2024; Goswami and Singh 2024). Such survey papers mainly provide a general overview of research developments in the imbalance learning field and highlight potential opportunities for future research. Other articles (Dou et al. 2021; Saini and Susan 2023; de Giorgio et al. 2023) discuss the trend and application of imbalance learning in specific domains. On the other hand, several articles (Leevy et al. 2018; Siers and Islam 2020; Tarawneh et al. 2022; Billion Polak et al. 2024) provide performance analysis from experimental studies by comparing the classification performance of various methods in binary class imbalance learning. Azhar et al. (2023) provided a more comprehensive performance analysis by examining the relationship between classification performance and data complexity, with a specific focus on the degree of overlap between classes. Similar studies were conducted by Vuttipittayamongkol et al. (2021) and Santos et al. (2022), which analyzed the impact of class overlap on binary class imbalance learning. Even though many review articles present comparisons of the classification performance of different methods, only a few provide baseline comparisons to assess the significance of the improvements achieved by each method. Such baseline comparisons were provided by Feng et al. (2021), Lin et al. (2023), Sauber-Cole and Khoshgoftaar (2022), and Rezvani and Wang (2023).

The many review and survey articles on binary class imbalance learning indicate that this problem has been extensively explored and analyzed from multiple perspectives. On the other hand, classification problems increasingly involve multiclass imbalanced datasets, as evidenced by the growing number of studies in this area. Table 1 lists six review and survey articles out of the 24 collected that specifically address multiclass imbalance learning. To highlight the distinctions and contributions of this review, we identify and compare several aspects discussed in these six prior articles: trend & application, performance evaluation, baseline comparison, difficulty factors, ad hoc methods, and dataset analysis. Based on this comparison, unexplored and unanalyzed research gaps remain. Therefore, our paper addresses these gaps by analyzing aspects of the proposed methods for handling multiclass imbalance problems that have yet to be thoroughly investigated.

Table 1 Comparison of several review articles on multiclass imbalance learning

Reference	Scope of review	Review contribution
Haixiang et al. (2017)	Trend & application	This paper provides a descriptive review of imbalanced learning, including data resampling, cost-sensitive, and ensemble learning. It presents general information on common approaches for addressing the multiclass imbalance issue and elaborates on various applications of imbalance learning.
Bi and Zhang (2018)	Performance evaluation. Baseline comparison	This paper provides an empirical study of earlier methods that used decompositional approaches. Most of the methods were published before 2010. This paper also proposes a new method to address multiclass imbalance learning, namely DECOC.
Johnson and Khoshgoftaar (2019)	Performance evaluation	This paper presents a critical review of various methods for addressing the class imbalance issue in deep learning. It presents performance evaluations gathered from previous works.
Tanha et al. (2020)	Performance evaluation	This paper focuses on the empirical study of various boosting methods for classification of multiclass imbalanced data. The study was conducted using several small and large datasets.
Lango and Stefanowski (2022)	Difficulty factor analysis. Dataset analysis	This paper presents an empirical study and a critical review of the factors that contribute to difficulty in multiclass imbalanced problems, which can lead to deterioration in classification performance. It highlights the primary issues that should be addressed when dealing with multiclass imbalanced datasets.
Yuan et al. (2023)	Performance evaluation. Difficulty factor	This paper presents a descriptive review of data resampling methods in the context of industrial data classification. It also provides a critical review of the limitations of the current state of the art and suggests several future research directions.
This review paper	Ad hoc method analysis. Performance evaluation. Baseline comparison. Difficulty factor analysis. Dataset analysis	This paper focuses on a critical review of recent ad hoc methods in data resampling and ensemble learning. It also conducts an empirical analysis to assess the effectiveness of both approaches in improving classification performance, as well as dataset analysis, covering class overlap degree and class configurations.

Up to this point, no review article has provided an in-depth analysis of the effectiveness of multiclass imbalance learning methods, even more regarding their relationship with various difficulty factors present in multiclass imbalanced datasets. Such an investigation is crucial to determine the extent of performance improvement achieved by existing methods. Moreover, gaining insights into optimal strategies for addressing these difficulty factors is essential. A proper understanding of handling difficulty factors can provide a clear direction for future research to improve and enhance the performance of existing methods.

Haixiang et al. (2017) discussed the trend and application of imbalance learning but did not provide more analytical information. The review focuses on where and when imbalance learning should be implemented, providing only general information related to imbalance learning methods. In contrast, the rest of the articles (Zhang et al. 2018; Tanha et al. 2020; Johnson and Khoshgoftaar 2019; Lango and Stefanowski 2022; Yuan et al. 2023) focus on discussing performance evaluation of several ad hoc methods without providing comprehensive review about trend and application in multiclass imbalance learning. The performance evaluation is obtained by conducting independent experiments or directly referencing classification performance results from source articles for comparison. However, only Johnson and Khoshgoftaar (2019) included a comparison to baseline performance, representing the performance achieved without applying imbalance handling methods. In their study, various boosting methods were evaluated on multiclass imbalanced datasets, with performance compared both among the methods and against the baseline. We believe that providing a baseline comparison is crucial, as it enables the measurement of each algorithm's effectiveness by revealing whether the methods genuinely improve classification performance

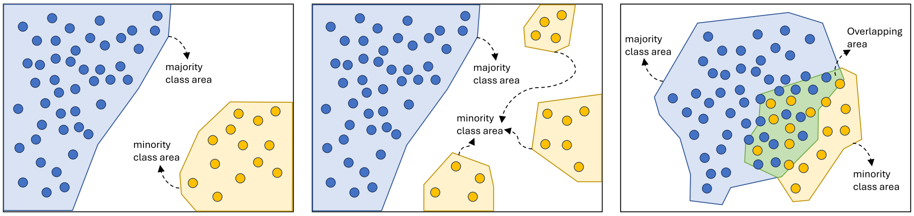
Furthermore, only a few articles discuss a deeper analysis of multiclass imbalance learning. Yuan et al. (2023) analyzed the performance evaluation and difficulty factors, including noise effects and class overlapping. Both factors require a specific approach to handle. Lango and Stefanowski (2022) offered an in-depth analysis of difficulty factors in multiclass imbalance learning, including class overlap, imbalance ratio, class configuration, and the number of classes. Their experiments reveal that class overlap is the most significant factor contributing to the deterioration of classification performance on imbalanced datasets. Additionally, specific class configurations and a higher number of classes also significantly impact the reduction of classification performance. Based on the analysis of difficulty factors, Lango and Stefanowski (2022) provided a dataset analysis that describes the percentage of overlap in benchmark datasets with respect to the class configuration of each dataset. This analysis offers insights into why certain datasets are more challenging to classify than others.

All reviews from previous studies are limited to discussion about decomposition approaches through OvO and OvA strategies, and most of them only provide general information without providing an in-depth review of each method. However, ad hoc methods tend to be more effective in handling multiclass imbalanced datasets, as noted by Janicka et al. (2019) and Wang and Yao (2012). Therefore, this paper provides an in-depth review of ad hoc methods, focusing on their performance evaluation, relationship to difficulty factors, baseline comparisons, and a comprehensive analysis of benchmark datasets. The review presented in this paper focuses on data resampling and ensemble learning approaches, as both are the most popular in multiclass imbalance learning.

3 Overview of imbalanced classification

Imbalance learning can occur in both classification and regression tasks. Particularly in classification, a dataset is considered imbalanced when one class has significantly more instances (the majority) than another (the minority). This condition can result from either natural causes or human errors. Natural causes arise when certain events or categories are inherently rare or difficult to observe, such as patients with specific diseases, credit card fraud, product defects, or system failures. On the other hand, human errors may lead to imbalance due to uneven data collection, accidental data loss, or mislabeling during the data annotation process. While imbalances caused by human errors can often be mitigated through additional data collection, this solution does not apply to naturally imbalanced scenarios. Therefore, computational approaches are necessary to address imbalanced datasets resulting from such natural causes.

However, imbalanced classification is challenging due to the uneven distribution of instances, especially in minority classes. As illustrated in Fig. 1, the instances in the minority class can be concentrated in a single region, but they may also be scattered across multiple distinct clusters. The small number of minority samples leaves a large undefined space in the feature space—regions where it is unclear whether they belong to the majority or the



(a) The minority class is concentrated in a single region.

(b) The minority class scatters across multiple distinct clusters.

(c) The minority class overlaps with the majority class.

Fig. 1 Illustration of several possible uneven data distributions in an imbalanced dataset. The blue circles represent instances from the majority class, and the yellow ones represent instances from the minority class

minority class. Therefore, this large, undefined space generates numerous possible decision boundaries created by machine learning algorithms, which can result in suboptimal decision boundaries between classes. These boundaries may achieve a low error rate on the training set but produce a high error on the testing set, as unseen instances in the testing set may fall outside the decision boundary of their true class. This situation can become even worse when many instances from the minority class overlap with those from the majority class, where some of the minority class instances may be misclassified as belonging to the majority class.

Generally, imbalanced classification can be addressed through common methods: data resampling, cost-sensitive learning, and ensemble learning. Data resampling aims to balance class distribution by either increasing the number of minority instances (oversampling) or reducing the number of majority instances (undersampling). Data resampling can be considered as part of data preprocessing as it is performed before training the classifier. Thus, it is independent of any classification algorithm. On the other hand, cost-sensitive learning assigns specific weights or costs to each class, thereby increasing the priority of the minority class in being recognized by the classifier. In contrast to data resampling, cost-sensitive learning is dependent on the classification algorithm used, requiring a different cost matrix for each algorithm. Furthermore, ensemble learning is another popular method in this field. It leverages the use of diverse classifiers to address the class imbalance issue. Apart from these common methods, a few studies have proposed alternative approaches to handle imbalanced classification, such as using the Hausdorff distance (Sevakula and Verma 2013, 2020). Hausdorff distance measures the closeness of a group to other groups, thereby helping to determine the boundary of each class (Sevakula and Verma 2013), which serves as the basis for data classification. However, this approach has been rarely used in recent methods to address the class imbalance issue.

Data resampling, however, still dominates the state-of-the-art methods to address the class imbalance issue. This is reasonable, as balancing the class distribution tends to be easier and more flexible than modifying specific algorithms to handle imbalanced datasets, whether through cost-sensitive learning or the use of specialized kernels. The synthetic minority oversampling technique (SMOTE) (Chawla et al. 2002) represents a breakthrough in oversampling methods, which previously were limited to random oversampling. SMOTE has become one of the most widely used techniques in addressing class imbalance. More than 85 methods now extend SMOTE to achieve better performance (Kovács, 2019). On the other hand, undersampling techniques, such as random undersampling (RUS), edited nearest neighbors, and TomekLink, are also widely used in many studies related to imbalanced classification. Despite the significant development of data resampling methods, they can be computationally expensive when applied to large datasets, as most approaches rely on nearest-neighbor search mechanisms.

Various ensemble learning approaches have been proposed by integrating data resampling into their mechanisms to further enhance classification performance. Some popular frameworks include SMOTEBagging (Wang and Yao 2009), RUSBagging (Wang and Yao 2009), RUSBoost (Seiffert et al. 2010), and EasyEnsemble (Liu et al. 2006). These ensemble frameworks offer advantages through the use of diverse classifiers, which can reduce either bias or variance, thereby improving performance. Ensemble learning is also flexible in employing various machine learning algorithms as base classifiers. Additionally, the aggregation method, which combines predictions from each base classifier, can be customized according to the specific problem at hand. These advantages within the ensemble learning framework make it a promising technique for addressing the class imbalance issue.

Studies in the field of imbalanced classification have also led to the development of several evaluation metrics that are more suitable than accuracy, e.g., average accuracy (AvAcc), geometric mean (GM), and class-balanced accuracy (CBA). These metrics are more sensitive to class imbalance issues, as they evaluate performance on a per-class basis rather than over all classes, thus providing a more representative assessment of model performance on imbalanced datasets. These performance metrics can reasonably be applied beyond imbalanced classification, as they offer valuable alternative evaluation into classifier performance.

4 Challenges in multiclass imbalance learning

Multiclass imbalance learning has more complex characteristics compared to binary class one. According to Lango and Stefanowski (2022), a multiclass imbalanced dataset has at least four difficulty factors that significantly influence classification performance. The most significant factor is the class overlap involving more than two classes. The small disjunct, defined as a small set of rare or exceptional instances from one class isolated in the region of another class, also increases the misclassification rate, especially in minority classes. Since there are more than two classes in a multiclass dataset, the class configuration also contributes to increasing the complexity of the classification task. Class configuration represents the composition of majority and minority classes within a dataset. Another difficulty factor is the number of classes. The following subsections detail the four difficulty factors in multiclass imbalance learning.

4.1 Multiclass overlap

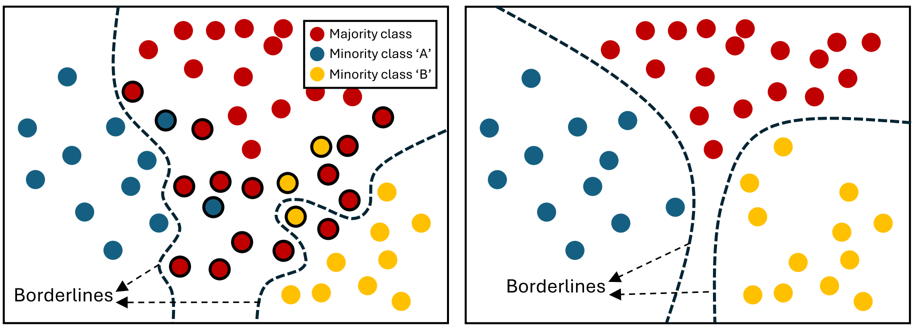
Conditions of multiclass overlapping generally occur in the border area between one class and the others (borderline)—the more complex the overlapping area of a dataset, the higher the misclassification rate. Vuttipittayamongkol et al. (2021) asserted that class overlap in a dataset has been an isolated challenge more difficult to deal with than the imbalance of the dataset itself. Moreover, they stated that if an imbalanced dataset has a low degree of overlap, the imbalance will have little impact on the classifier's performance.

The existence of class overlap can increase the level of difficulty in separating data between classes. Additionally, class overlap can affect or complicate the classifier's determination of the best borderline to separate the data between classes. As illustrated in Fig. 2a, the formed borderlines may strictly follow the data distribution patterns in the boundary area due to the presence of class overlap. This potentially results in an overfitted classification model. If the overlapping instances from the majority class are eliminated, as shown in Fig. 2b, the classifier can more easily find the optimal borderlines separating the classes, resulting in a model with better generalization ability.

The class overlap can be addressed using an undersampling technique by reducing the number of instances in the majority classes. According to Vuttipittayamongkol et al. (2021), an effective technique for addressing class overlap in datasets is to utilize overlap-based resampling techniques, which can eliminate instances from the majority class that are located in the borderline area with instances of the minority class. This technique is even recommended for addressing class overlap in a balanced dataset. On the other hand, although not as popular as resampling techniques, a kernel-based learning approach can also be an alternative method for solving problems on datasets that contain class overlap, as demonstrated by Wang et al. (2021). The technique works by mapping data into a feature space so that a hyperplane formed by a kernel can separate data distribution between classes. This technique typically employs a support vector machine (SVM) algorithm, which can effectively classify non-linear data. However, obtaining an optimal hyperplane requires the selection of kernel functions and correct hyperparameter settings (Mathew et al. 2018).

4.2 Small disjunct and noise

Similar to the class overlap factor, the existence of small disjunct also tends to have a significant negative impact on classification performance. Based on the research of Jo and Japkowicz (2004), the treatment of imbalanced datasets is effective when dealing with the presence of small disjunct. In many cases, small disjunct is usually considered label noise (Koziarski et al. 2020) due to the possibility of mislabeling when preparing the dataset. However, small disjunct is fundamentally different from noise. A small disjunct is typically a small group of instances with a specific pattern, originating from instances of one



(a) Borderlines in dataset with class overlap

(b) Borderlines in dataset without class overlap

Fig. 2 Illustration of possible decision boundaries formed on a dataset with **a** multiclass overlap and **b** without class overlap. Circles with black outlines indicate overlapping instances class that may share very similar features with those of another class. Meanwhile, noise is typically a rare, random instance without a specific pattern, often resulting from mislabeled instances or incorrect values in the features (Salgado et al. 2016).

Azhar et al. (2023) remarked that one way to deal with small disjunct is to use a cluster-based resampling technique that groups instances on each class into several clusters, then performs resampling technique by generating a number of synthetic instances on every such cluster (oversampling) or otherwise deleting a certain amount of instances from each cluster (undersampling). Nevertheless, the cluster formation process must use appropriate algorithms and parameter settings to ensure the clusters are formed optimally.

Liu et al. (2024) proposed a label correction on noisy samples based on a semi-supervised learning approach. The corrected labels are then incorporated into the training process. Semi-supervised learning has also been successfully employed using a pseudo-labeling mechanism, where unlabeled instances are assigned labels based on a confidence threshold to ensure only confidently predicted samples are included (Zhu et al. 2024). Furthermore, Miao et al. (2024) combined contrastive semi-supervised learning with collaborative sample selection to address the influence of noisy instances in the dataset. However, to achieve optimum results in semi-supervised learning, the labeled samples in the training set must be of high quality and free from noise.

Another approach uses the intuitionistic fuzzy method to handle noise within the dataset. The nature of fuzzy systems, which can measure the degree of membership of an object in a certain domain, is beneficial for detecting noise and distinguishing it from clean samples (Chen et al. 2025). This approach has been effectively employed by Guo et al. (2024) to assign weights to each sample, thereby mitigating the impact of noise. Intuitionistic fuzzy methods also help enhance the performance of random forest (RF) by applying intuitionistic fuzzy information gain to guide feature selection while incorporating hesitation degrees during the information transfer process (Ren et al. 2023).

4.3 Class configuration

In multiclass imbalance learning, the class configuration becomes a difficulty factor that requires special handling that cannot be effectively solved using algorithms designed for binary class datasets. Wang and Yao (2012) divided the class configuration into two categories, multi-majority and multi-minority. A multi-majority dataset has many majority classes but only one minority class. On the contrary, a multi-minority configuration has many minority classes and only one majority. Buda et al. (2018) defined another category of class configuration, a linear or gradual imbalance, in which the amount of instances in each class increases linearly.

According to Lango and Stefanowski (2022), the multi-majority configuration is the most difficult to handle because the majority class dominates the datasets. It is further stated that the overlap between the majority and the minority classes leads to a more significant decrease in performance on the classifier than between the corresponding minority classes. Figure 3 illustrates the differences between the three categories of class configuration on the multiclass imbalance problem.

However, there is still no formal definition for categorizing majority and minority classes in multiclass imbalance problems. Grina et al. (2022) suggested using a threshold based on the average size of all classes in a dataset to categorize majority and minority classes. A

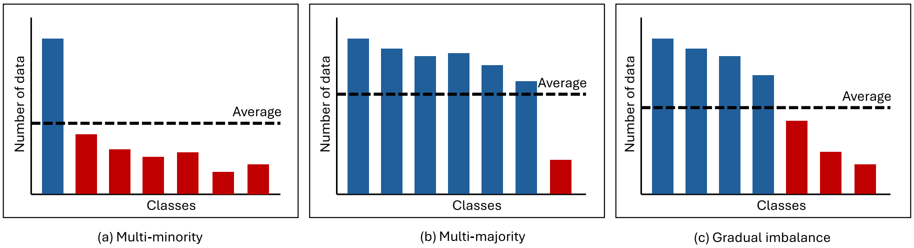


Fig. 3 Illustration of the class configuration of a dataset containing seven classes: **a** multi-minority, **b** multi-majority, and **c** gradual imbalance. The blue and red bars represent the majority and minority classes, respectively. The dotted lines represent the average size of all classes

class with more instances than the threshold is considered a majority class; otherwise, it is regarded a minority class. Alternatively, the concept of the imbalance ratio used in binary classification can be applied, where a class is considered a minority if its imbalance ratio exceeds 1.5 (Yang et al. 2018); otherwise, it is classified as a majority class. The imbalance ratio of a class is the ratio of the number of instances in the largest class to the number of instances in the class in the dataset.

Furthermore, multiclass datasets have a concept called relative majority (and relative minority) (Zheng et al. 2022), which is a condition in which a class can be a majority when compared to a much smaller class, but at the same time, can also be a minority if compared to other larger-sized classes (and vice versa) (Krawczyk 2016b; Liu et al. 2021). This concept can be considered when applying data resampling to a dataset with multiclass imbalance.

4.4 Number of classes

Apart from the class configuration factor, the number of classes in a dataset also becomes another difficulty factor in the multiclass imbalance problem. A large number of classes in a dataset will reduce the recall value of a minority class (Lango and Stefanowski 2022). This is because the larger the number of classes, the greater the opportunity for overlap to occur, so it is directly proportional to the potential for data misclassification. However, it is impossible to manipulate the number of classes in the dataset; therefore, the opportunities for method improvement can only be focused on the three complicating factors described previously, i.e., class overlap, small disjunct, and class configuration.

5 State-of-the-art in multiclass imbalance learning

In general, there are two main approaches to handling imbalanced datasets, i.e., data-level and algorithm-level approaches, as depicted in Fig. 4. The data-level approach balances the size of classes within the dataset by performing data resampling, i.e., oversampling, undersampling, or both. This can help the classifier learn the pattern of each class in balance and reduce the learning bias. According to Haixiang et al. (2017) and Kaur et al. (2019), feature engineering, including feature selection and extraction, can also be part of the data-level approach. It can help modify and optimize the dataset by selecting a subset of significant

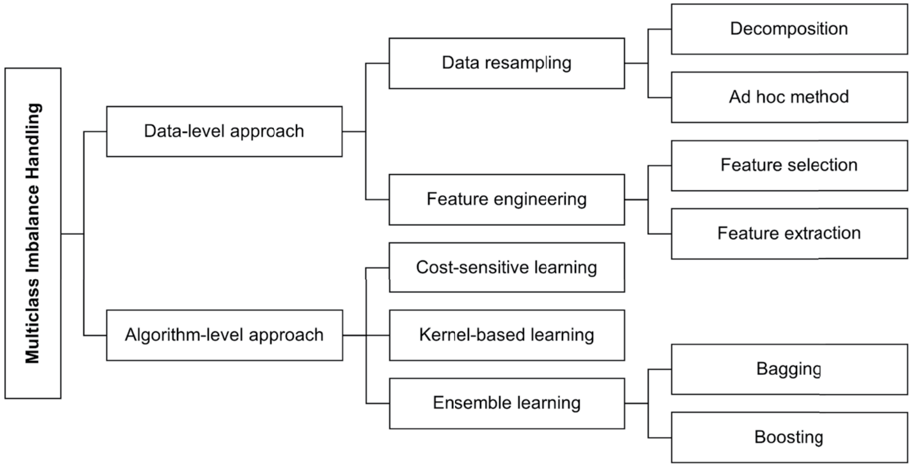


Fig. 4 Taxonomy of methods for handling classification on multiclass imbalanced datasets. The most common methods are data resampling and ensemble learning

features or extracting new features using specific algorithms, such as principal component analysis (PCA). However, feature selection and extraction have been less used in previous works than the resampling strategy (Haixiang et al. 2017). Since the data-level approach is effective in the data preprocessing phase, it can be applied universally to any classifier.

On the other hand, the algorithm-level approach improves the classifier to learn from the imbalanced dataset, conducted by configuring the class weight parameter using a cost-sensitive learning (Krawczyk 2016a; Wang et al. 2019) mechanism or adding a particular kernel to modify the hyperplane that separates the classes (Chen et al. 2022; Ding et al. 2018; Raghuwanshi and Shukla 2019). In contrast to the data-level approach, both methods work based on specific classification algorithms. However, configuring the optimal cost learning is challenging in cost-sensitive learning as it requires specific observations or expert knowledge (Krawczyk 2016b).

Another strategy that emerges at the algorithm level is the use of ensemble learning. Among algorithm-level approaches, ensemble learning is the most popular in multiclass imbalance learning (Tanha et al. 2020). Additionally, ensemble learning works independently of any classification algorithm, similar to data-level approaches. Combining ensemble and data-level approaches can improve the classification performance (Werner de Vargas et al. 2023). This paper reviews the state of the art in data resampling and ensemble learning, the most popular methods. The review focuses solely on the ad hoc methods rather than decomposition in the data resampling approach.

5.1 Data resampling approach

Data resampling in multiclass imbalance learning can be conducted in two ways: adapting binary class resampling through a decomposition strategy and developing an ad hoc method to deal with multiclass imbalanced datasets. One-versus-one (OvO) and one-versus-all (OvA) are the most popular decomposition methods. OvO decomposes a dataset containing N classes into $\frac{N(N-1)}{2}$ subsets. In every subset, each class is paired with another class.

Meanwhile, OvA decomposes a dataset with N classes into N subsets, where each subset contains a pair of one class and a combination of other classes. Fernández et al. (2013) and Li et al. (2020) reported that their experiments suggested OvO tends to be more effective than OvA. However, the decomposition approach is less effective than the ad hoc method, which is specifically designed to handle multiclass imbalanced datasets (Janicka et al. 2019; Wang and Yao 2012).

Tables 2 and 3 list 14 ad hoc methods in data resampling that have been proposed from 2011 until 2024. This amount is significantly less than data resampling methods for binary class imbalance. Kovács, 2019 revealed that there are 85 variants of oversampling methods for handling binary class imbalanced datasets. Most of these methods are extensions of the well-known SMOTE algorithm first introduced by Chawla et al. (2002). SMOTE itself is used in several ad hoc methods for multiclass imbalance learning. It generates synthetic instances based on interpolation between two adjacent instances.

This paper discusses the data resampling approaches based on the resampling scope, step, strategies, and selection method. It provides readers with a comprehensive perspective on resampling mechanisms used in multiclass imbalance learning. Moreover, this can help recognize which resampling mechanisms will work effectively on various issues.

5.1.1 Resampling scope

The resampling scope is defined as the area in which resampling is performed. Based on the ad hoc methods in Tables 2 and 3, the resampling scope can be divided into local and global. As illustrated in Fig. 5b, local resampling is conducted within the local boundary of an observed instance or neighborhood area. In contrast, global resampling uses all instances in a class to calculate the class centroid, which serves as a reference for generating synthetic instances or removing existing ones, as shown in Fig. 5c.

The SMOTE-based oversampling methods, i.e., S-SMOTE (Fernández-Navarro et al. 2011), SCUT (Agrawal et al. 2015), SMOM (Zhu et al. 2017), and NROMM (Liu et al. 2023), work locally because it needs to identify the neighborhood to generate synthetic instances. Some boundary-based oversampling methods, e.g., MC-RBO (Krawczyk et al. 2020), MC-CCR (Koziarski et al. 2020), SHSampler (Zheng et al. 2022), OREM-M (Zhu et al. 2023), and MC-NRO (Shen et al. 2024), also work locally based on the neighborhood of the observed instance. On the other hand, some boundary-based resampling methods, such as MDO and OCSV-US, operate globally based on the data distribution within each class. Following Abdi and Hashemi (2016) and Han et al. (2023), global resampling uses the Mahalanobis distance instead of the Euclidean distance to determine the position of synthetic instances. The Mahalanobis distance assumes that each feature correlates with the others, producing synthetic instances that are more representative than those produced by the Euclidean distance (Abdi and Hashemi 2015).

Local resampling has the advantage of using the local distribution from the observed instance, which can help the algorithm generate more representative synthetic instances and handle the small disjunct problem. However, local resampling may be more computationally intensive than global resampling, as it must calculate the distance from each instance to the others. On the contrary, global resampling tends to be more efficient because the resampling mechanism only involves the distance of each instance to the centroid of its own class. Unfortunately, determining a class's center point can produce bias if a class is distributed into several clusters. Furthermore, global resampling also has the potential to exacerbate the class overlap issue in the dataset, as it does not account for the presence of other classes when generating synthetic instances.

Other methods attempt to combine global and local resampling (hybrid) to leverage the advantages of both strategies. GLOS, introduced by Han et al. (2023), employs several resampling techniques that operate both locally and globally. This method utilizes an MMO algorithm that operates in the global scope, adapted from MDO. Within the local scope, it employs two different oversampling algorithms, namely AMBO and RRO, which operate by generating synthetic instances within a spherical area of a specified radius.

5.1.2 Resampling step

The resampling step is defined as the number of resampling techniques used in an ad hoc method. The concept of hybrid resampling, which involves the sequential use of oversampling and undersampling to balance class distribution, is a key focus in this case. It is important to note that GLOS, a method that combines global and local resampling, is not considered a hybrid resampling despite its use of multiple resampling techniques. This is because GLOS only performs oversampling on the imbalanced dataset.

Hybrid resampling typically occurs when a dataset has an extreme imbalance ratio or aims to minimize class overlap within the dataset. Otherwise, a single resampling technique, such as oversampling or undersampling, can be applied individually to the dataset. Generally, single resampling typically performs oversampling rather than undersampling. However, OCSV-US (Krawczyk et al. 2021) presents a different strategy by only undersampling the imbalanced dataset. This method considers that some instances have a more important role than others. Therefore, it utilizes SVM to identify the support vectors of each class through one-class classification (Alam et al. 2020) and removes the remaining instances using evolutionary undersampling

Most existing methods in Tables 2 and 3 operate in the hybrid schema by performing oversampling before undersampling or vice versa to minimize the effect of multiclass overlap, which can deteriorate classification performance. This hybrid resampling is performed by oversampling the minority classes and undersampling the majority classes until each reaches a certain threshold. Janicka et al. (2019) defined the threshold as the average size of the smallest majority class and the largest minority class within the dataset. Other works (Agrawal et al. 2015; Grina et al. 2022; Zheng et al. 2022) defined it as the average size of all classes.

In the context of hybrid resampling, the sequence of resampling techniques is crucial for achieving optimal results. When conducted first, oversampling can enhance the representation of minority classes in the dataset. However, this approach can lead to an increase in the dataset's size, thereby escalating the computational time required for undersampling. Additionally, in cases of extreme imbalance ratio, the introduction of synthetic instances can potentially result in overfitting. On the other hand, performing oversampling after undersampling can reduce computational time due to the smaller dataset size resulting from post-undersampling. Furthermore, this sequence can also help prevent the classifier from overfitting to the synthetic instances.

Table 2 Summary of ad hoc data resampling methods in multiclass imbalance learning

Method	Year	Resampling mechanisms					Method's workflow
		Scope	Oversampling	Undersampling	Class handling	Instance selection	
S-SMOTE	2011	Local	SMOTE-based	–	Per-class	None	SMOTE is applied to each minority class, and the augmented dataset is then used to train an RBF neural network optimized using a memetic algorithm.
SCUT	2015	Local	SMOTE-based	Cluster-based	Per-class	None	SCUT performs clustering on each majority class and randomly removes instances from each cluster. Next, SMOTE is then applied in each minority class.
MDO	2016	Global	Boundary-based	–	Per-class	Weighted random	MDO randomly selects instances from the minority classes based on their weights, transforms them into principal component (PC) space, and then generates synthetic instances within the PC space.
SMOM	2017	Local	SMOTE-based	–	All classes	None	SMOM first clusters each class and assigns a cluster label of either 'safe' or 'trapped' to each instance. Then, for each instance in the minority class, synthetic instances are generated based on the cluster labels of its nearest neighbors from the entire dataset.
SOUP	2019	Local	Duplication-based	Priority-based	Per-class	Priority-based	SOUP calculates a safe level for each instance. Instances in the majority classes with low safe levels are removed, while instances in the minority classes with high safe levels are duplicated.
MC-RBO	2020	Local	Boundary-based	–	Pairwise	Random	MC-RBO calculates the potential area around each minority instance using a Gaussian radial basis function (RBF). Minority instances are randomly selected, and synthetic instances are then generated around each selected instance following the Gaussian RBF distribution.
MC-CCR	2020	Local	Boundary-based	Data translation	Pairwise	Random	MC-CCR translates majority instances away from nearby minority instances. It then creates a spherical boundary around each randomly selected minority instance and generates random synthetic instances within this boundary.

Table 2 (continued)

Method	Year	Resampling mechanisms					Method's workflow
		Scope	Oversampling	Undersampling	Class handling	Instance selection	
OCSV-US	2021	Global	–	Evolutionary-based	Per-class	Priority-based	OCSV-US identifies the optimal hyperplane that separates the classes and removes all instances in the dataset except those that serve as support vectors for each class. A genetic algorithm is then used to select the most significant support vectors and discard the rest.

Table 3 Summary of ad hoc data resampling methods in multiclass imbalance learning (continued from Table 2)

Method	Year	Resampling mechanisms					Method's workflow
		Scope	Oversampling	Undersampling	Class handling	Instance selection	
SHSampler	2022	Local	Boundary-based	Probability-based	All classes	Weighted random	SHSampler calculates the safe level of each instance in the dataset. Instances in the majority classes with low safe levels are randomly selected for removal. In contrast, instances in the minority classes with high safe levels are randomly selected to serve as the basis for generating synthetic instances within a spherical area around them.
MC-EVHS	2022	Global	SMOTE-based	Priority-based	All classes	Priority-based	MC-EVHS assigns evidential membership to each instance based on the Dempster-Shafer theory. Instances located near the class center have higher evidential membership than those farther away. Oversampling is performed on instances with lower evidential membership using an interpolation approach similar to SMOTE.
NROMM	2023	Local	SMOTE-based*	Cluster-based	Pairwise	Priority-based	NROMM clusters the instances in each class into three categories: inland, borderline, and trapped. It then applies different oversampling strategies to minority instances in each category and removes borderline and trapped instances from the majority classes.
OREM-M	2023	Local	Boundary-based	-	All classes	Random	OREM-M iteratively explores the potential area of each minority class from the entire dataset. For each potential area of an observed minority instance, OREM-M identifies all possible clean sub-regions that do not contain any instances from other classes. Synthetic instances are then generated within these sub-regions.

Table 3 (continued)

Method	Year	Resampling mechanisms					Method's workflow
		Scope	Oversampling	Undersampling	Class handling	Instance selection	
GLOS	2023	Hybrid	Boundary-based	-	All classes	Priority-based	GLOS classifies each instance as safe, borderline, or small disjunct. It applies MDO-based oversampling to safe instances, SMOTE-borderline to borderline ones, and radial-based oversampling to small disjuncts.
MC-NRO	2024	Local	Boundary-based	Data translation	Pairwise	Random	MC-NRO combines the strategies of MC-CCR and MC-RBO. It first translates majority instances away from nearby minority instances and then applies radial-based oversampling to randomly selected minority instances. Finally, it removes translated majority instances that still overlap with other classes.

*The authors did not mention the oversampling method explicitly, instead just stated "Interpolate to generate S_i and S_b." Hence, in this paper it is categorized as SMOTE-based to simplify the explanation

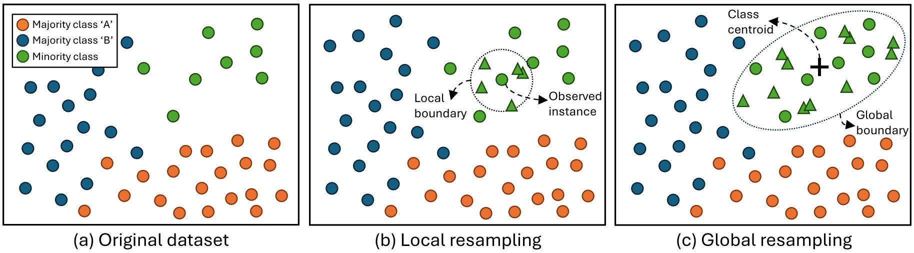


Fig. 5 Illustration of **a** a multiclass imbalanced dataset, resampled using **b** local resampling and **c** global resampling. Triangles represent the synthetic instances

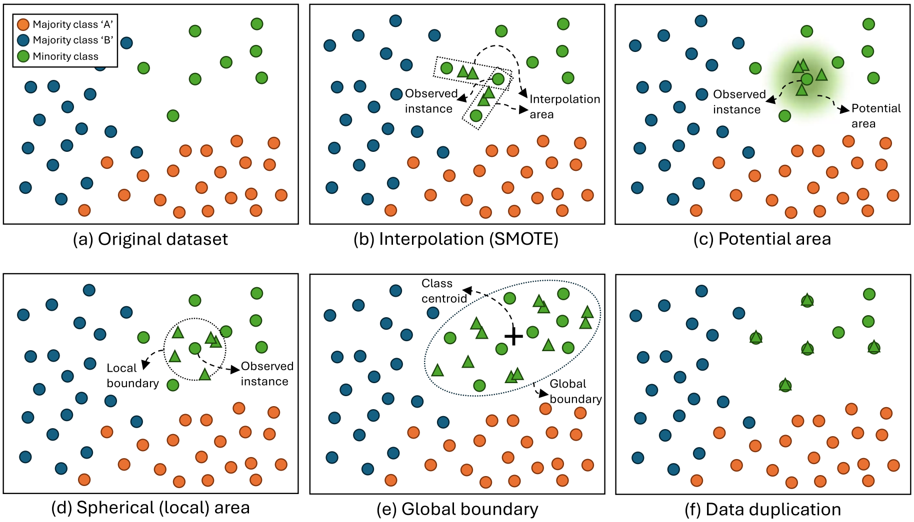


Fig. 6 Illustration of oversampling mechanisms in multiclass imbalance learning. Triangles represent synthetic instances

5.1.3 Oversampling mechanisms

The oversampling mechanism represents the procedure for carrying out oversampling within the minority class. This part has the most important role in producing optimally balanced datasets. Based on the previous works in Tables 2 and 3, oversampling mechanisms can be divided into three categories: SMOTE-based, boundary-based, and duplication-based. The boundary-based itself has three different oversampling strategies, i.e., based on potential area, spherical area, and global boundary.

5.1.3.1 SMOTE-based oversampling SMOTE is the earliest oversampling method that introduces the generation of synthetic instances by interpolating between two adjacent instances. As illustrated in Fig. 6b, the observed instance takes two of its nearest neighbors and generates random instances between each pair. This mechanism is used in S-SMOTE, SCUT, SMOM, MC-EVHS, and NROMM. SMOTE has the significant drawback of producing noisy instances that can worsen the class overlap. Therefore, additional mechanisms should be implemented to deal with this issue. SCUT and NROMM utilize clustering algorithms to group instances into clusters and perform undersampling within each cluster before oversampling.

SCUT performs oversampling using SMOTE directly to every minority class. Meanwhile, MC-EVHS first identifies each instance, whether it is an overlapping sample, label noise, or an outlier. Then, SMOTE is performed on instances in one of those three categories. MC-EVHS aims to strengthen the presence of minority classes in border areas. This mechanism serves the same purpose as Borderline-SMOTE (Han et al. 2005). Like MC-EVHS, NROMM first categorizes the instances in each cluster as inland, borderline, or trapped based on their neighborhood. The oversampling is performed using SMOTE for each group. A specific procedure, namely adaptive embedding, is employed in the trapped group to help SMOTE generate instances more safely. However, NROMM has more complex computations than SCUT and MC-EVHS.

Another method, SMOM, employs a clustering algorithm to generate multiple clusters for each minority class. It calculates the selection weight for each instance in every cluster to rank the oversampling order. While other methods group the instances into clusters or identify each instance in the dataset, S-SMOTE focuses on optimizing the classifier, i.e., radial basis function neural networks (RBFNN), using a memetic algorithm. It optimizes the network architecture and weights of RBFNN. In this method, SMOTE is performed in each minority class before the learning process using the memetic algorithm and RBFNN.

5.1.3.2 Boundary-based oversampling One of the most important problems in multiclass imbalance learning is class overlap. This factor has an even more substantial impact when a dataset contains many majority classes, i.e., a multi-majority dataset (Lango and Stefanowski 2022). Some studies deemed interpolation-based oversampling, such as SMOTE, ineffective in dealing with this issue. Therefore, several previous works proposed boundary-based oversampling techniques to minimize the impact of class overlap on classification performance. Tables 2 and 3 list several such methods. Some use a local boundary, i.e., MC-RBO, MC-CCR, SHSampler, OREM-M, and MC-NRO. Another method uses a global boundary as in MDO, and the rest combines both global and local boundaries as in GLOS.

MC-RBO uses the potential area around the observed instance using Gaussian radial basis function (RBF) distribution. As illustrated in Fig. 6c, the potential area has the highest probability near the observed instance, indicating that the synthetic instances are most likely generated in his area. This probability decreases as the distance increases. The coverage of the potential area can be adjusted by changing the spread parameter in the Gaussian RBF. This strategy can reduce the prospect of the algorithm generating noise that can worsen the class overlap. Moreover, the local boundary also helps solve the small disjunct problem within the dataset. However, experimentation is needed to find the optimal value of the spread parameter.

On the other hand, MC-CCR, SHSampler, and MC-NRO use the spherical area to generate synthetic instances around the observed instance. As depicted in Fig. 6d, the observed instance serves as the center of the spherical area, with its size determined by a specified radius. In MC-CCR, the radius is user-defined, while in SHSampler and MC-NRO, it is determined by the distance of the observed instance to the *k*-th nearest neighbor from the other class. Unlike the potential area in MC-RBO, all points within the spherical area have an equal probability of generating synthetic instances.

Similar to MC-RBO, MC-CCR also helps handle the small disjunct problem and minimize the possibility of creating noise. However, the user-defined radius in MC-CCR requires experimentation to determine the optimal value. Meanwhile, SHSampler has a procedure to determine whether each instance is in a safe area. This involves the center point of a class, which can be biased if a class is divided into several clusters.

OREM-M also uses a spherical area with a radius based on the distance of two adjacent instances from the same class. It has a two-step boundary-searching procedure. The first step takes one from the nearest neighbors of an observed instance and then creates a spherical area based on the distance between the two instances as the radius. The second step searches for subregions within the preformed spherical area to create another small spherical area. This subregion is expected to be clean of other class instances for oversampling. However, OREM-M lacks a data cleaning mechanism, unlike MC-CCR and SHSampler. Therefore, when generating a synthetic instance in a subregion, it must ensure that this instance is not in the neighborhood of any instances from other classes. This mechanism may require higher computational time since it is conducted iteratively.

Meanwhile, MDO performs oversampling within the global area of a minority class. Figure 6e depicts that the class centroid is used as the basis of reference to generate synthetic instances in the area of the class. MDO uses Mahalanobis distance instead of Euclidean distance to determine the area for generating synthetic instances since it enables MDO to generate synthetic instances that align with the tendencies of the minority class distribution. This ensures that the synthetic instances adhere to the nature of the original sample. A similar strategy is also adopted in GLOS to perform oversampling on instances in safe areas. However, using this oversampling strategy may introduce noise and increase the risk of class overlap, as the synthetic instances generation process does not account for the presence of instances from other classes.

5.1.3.3 Duplication-based oversampling Data duplication is the fundamental oversampling technique widely used in earlier works on imbalance learning. As illustrated in Fig. 6f, several instances from the minority class are randomly duplicated to match the size of the majority class. Among the methods in Tables 2 and 3, only SOUP (Janicka et al. 2019) uses this duplication-based oversampling. However, random duplication may lead to overfitting condition. Therefore, SOUP has additional operations to avoid overfitting by prioritizing the oversampling based on the safe level of the observed instance. Instances with the highest safe level will be duplicated first. The duplication mechanism in SOUP effectively avoids generating noise, but it cannot enhance the presence and enlarge the distribution area of the minority class.

5.1.4 Undersampling mechanisms

Undersampling, especially in multiclass imbalance learning, significantly minimizes the impact of class overlap and helps reveal the existence of small disjuncts. Various undersampling methods have been proposed in earlier studies. Based on Tables 2 and 3, six different undersampling mechanisms are used in the ad hoc method in multiclass imbalance learning.

5.1.4.1 Cluster-based undersampling The primary purpose of cluster-based undersampling is to keep the representation of all sub-concepts within a class (Agrawal et al. 2015) by removing instances from the majority class while maintaining the data distribution. According to Wang et al. (2020), cluster-based undersampling is much better than randomly removing instances using random undersampling. As illustrated in Fig. 7b, cluster-based undersampling groups instances from each class into several clusters using clustering algorithms. Undersampling is performed in each cluster using random undersampling (RUS) or other advanced undersampling methods for better results.

SCUT uses the expectation maximization algorithm to group instances in the majority classes into several clusters. This algorithm does not need user-defined input to determine the number of generated clusters (Agrawal et al. 2015). RUS is applied to each majority class when the clusters are formed. However, the randomness of RUS may still affect the change in class distribution (Tsai et al. 2019). On the other hand, NROMM employs the CFSFDP clustering algorithm introduced by Rodriguez and Laio (2014) to categorize the instances into three groups: inland, borderline, and trapped.

Different from SCUT, NROMM performs clustering into majority and minority classes. Once the groups of instances are formed, undersampling is performed on the borderline and trapped groups by removing all samples inside them. Using this strategy, NROMM effectively reduces the size of the majority class while maintaining the class distribution. However, to achieve this result, NROMM requires expensive computation. Additionally, NROMM remains ineffective on datasets containing numerous borderline samples (Liu et al. 2023).

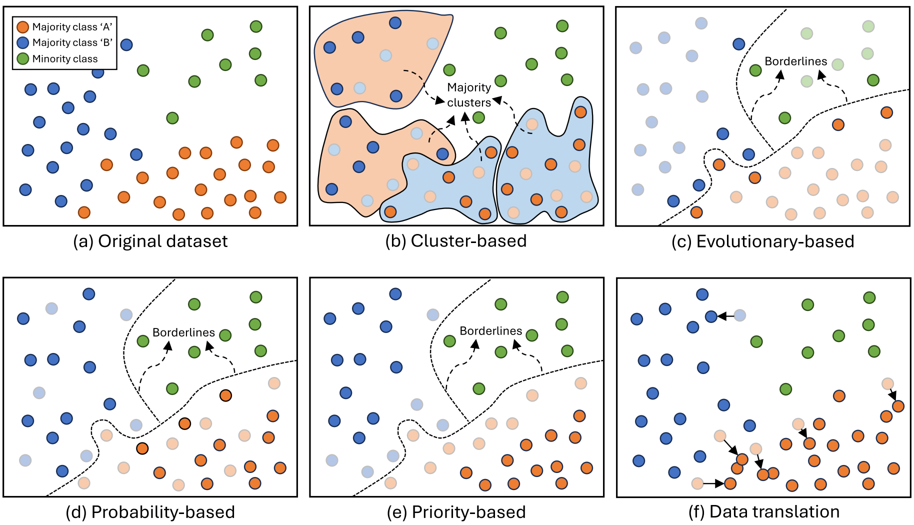


Fig. 7 Illustration of undersampling mechanisms in multiclass imbalance learning. Transparent circles represent the deleted instances from the majority classes

5.1.4.2 Evolutionary-based undersampling Optimization algorithms, such as genetic and other evolutionary algorithms, are suitable for undersampling mechanisms. As illustrated in Fig. 7c, evolutionary algorithms select important instances, e.g., those near the decision boundaries, and eliminate the rest. Triguero et al. (2017) introduced this approach with evolutionary undersampling (EUS). In the case of multiclass imbalance learning, EUS is used in OCSV-US (Krawczyk et al. 2021), which combines SVM and genetic algorithm to perform undersampling on the imbalanced dataset.

OCSV-US proceeds by first doing one-class classification using SVM to determine the support vectors from each class (including the minority class). After that, the genetic algorithm performs combinatorial optimization to select the most significant samples to improve the classification performance of the imbalanced dataset. This two-step mechanism occurs in all majority and minority classes. Therefore, this method produces a small dataset containing only the optimal support vectors. However, SVM and genetic algorithms require fine-tuned parameters to perform effectively. In addition, when working with a large dataset containing many classes, OCSV-US may require a higher computational time due to the increased search space of the genetic algorithm.

5.1.4.3 Priority-based undersampling Removing instances from the majority class based on certain priorities can help eliminate unimportant samples from the dataset. This approach aims to reduce the presence of the majority class in borderline areas. The priority order can be established using a statistical approach. Based on Fig. 7e, instances located far from the borderlines are assigned a higher priority and are thus retained in the dataset. This helps to reduce overlap around the borderlines.

As in SOUP, the priority order is determined by the safe level of each observed instance. An observed instance will have a higher safe level if surrounded by more instances from the same class than other classes. If all neighbors of an observed instance are from the same class, the safe level is 1, and 0 if none of the observed instance's neighbors are from the same class. Between these two extremes, the safe level ranges from 0 and 1. The undersampling mechanism in SOUP is carried out in ascending order, starting from the instances with the lowest safe level. Another method, MC-EVHS, proposed a soft evidential label for each instance set in the majority classes, calculated based on the Mahalanobis distance from each instance to the center of its class. This strategy adds a membership function to each instance class in the dataset. The closer the instance is to a class's centroid, the higher its evidential label towards that class.

5.1.4.4 Probability-based undersampling Probability-based undersampling is somewhat similar to priority-based undersampling. The key point of these two algorithms lies in data weighting, which represents the significance of each instance type for classification. Instead of ranking the instances based on their weights, probability-based undersampling uses the weights as the probability to select which instances to remove. The probability is inversely proportional to the weight; therefore, the lower the weight, the higher the probability that instances will be deleted. In prior studies, instances located in safe areas far from the borderlines are assigned lower weights, thus having a higher probability of being selected. As illustrated in Fig. 7d, most of the instances retained in the dataset are located in safe areas, which are far from the borderlines. However, one could assume the opposite by assigning higher probabilities to instances located near the borderlines.

SHSampler (Zheng et al. 2022) employs a probability-based approach to undersampling within the majority classes. A roulette wheel mechanism randomly selects the instances to be deleted based on their weight. This undersampling process is carried out until the size of the majority class matches the average size of all classes in the dataset. However, this approach can still retain insignificant instances in the dataset, which may affect classification performance.

5.1.4.5 Data translation undersampling This review article considers data translation as pseudo-undersampling because it does not reduce the number of instances in the majority classes. Based on the illustration in Fig. 7f, each majority instance located near other classes is translated in a certain direction to avoid overlap. Thus, this approach also helps maintain data distribution and prevent information loss.

Both MC-CCR and MC-NRO use this strategy to minimize the impact of class overlap between classes to improve classification performance. This method iteratively picks a pair of classes and performs data translation to the larger class. In MC-CCR, the translated instances may overlap with those of other classes because resampling is performed alternately between two classes. MC-NRO addresses this issue by removing previously translated instances in the second undersampling step if they overlap with instances of other classes.

5.1.5 Class Handling

Class handling defines how resampling methods organize multiple classes before applying oversampling or undersampling. Based on the recent studies in data resampling, the class handling for multiclass imbalance learning can be categorized into three types: per-class, pairwise, and all classes. Earlier methods, such as S-SMOTE, SCUT, MDO, and SOUP, employ per-class handling, where multiple classes are separated, and data resampling is performed individually for each class. While this strategy is computationally efficient, it suffers from the drawback of producing overlapped instances during oversampling, mainly because it ignores the existence of other classes.

On the other hand, the latest studies prefer using all classes and pairwise handling instead of separating the classes individually. Methods such as SHSampler, MC-EVHS, OREM-M, and GLOS originally perform resampling for each class by considering all classes at once, allowing them to have complete information about the data distribution across the entire dataset. While this approach is straightforward and exhaustive, it can be computationally expensive, particularly when working with large datasets. Moreover, this approach may also reduce the diversity of synthetic instances and make them less explorative due to the numerous constraints arising from involving all classes in the resampling process.

Other recent methods, including MC-RBO, MC-CCR, NROMM, and MC-NRO, employ an alternative approach by decomposing multiple classes into pairwise subsets and conducting resampling similar to that used in binary-class datasets. It is essential to note that this strategy differs from OvO or OvA mechanisms. Instead of aggregating predictions from each subset, as in OvO and OvA, this pairwise handling simply merges the resampled subsets into a single new dataset. In the methods mentioned above, decomposition is carried out gradually by iteratively pairing each class with larger classes. This approach can be less computationally expensive than handling all classes simultaneously, and it can also generate more diverse synthetic instances while still considering the presence of instances from other classes. Based on our experience, handling all classes together or using pairwise approaches is better than per-class handling, as both approaches take into account information from other classes during data resampling.

5.1.6 Resampling selection

Resampling selection is the method of selecting which instances from each minority class will be observed for oversampling or undersampling. Earlier methods, such as S-SMOTE, SCUT, SMOM, and MDO, apply the oversampling to all instances in each minority class and do not use any selection method to determine or prioritize instances to be resampled. However, some instances in a particular area may be more necessary to observe and resample than others, and there is no need to resample all instances within the minority classes. Therefore, later resampling methods propose specific mechanisms to select significant instances within the minority class for observation and resampling.

Several resampling methods listed in Tables 2 and 3 use a priority-based approach to resampling selection, i.e., SOUP, OCSV-US, MC-EVHS, NROMM, and GLOS. This approach focuses on only the important instances that have a substantial impact on enhancing the presence of the minority class in relation to the majority class. The priority can be represented by adding weight to each instance or segmenting the instances into several categories.

SOUP and MC-EVHS calculate the weight based on the instance's difficulty level. Both methods consider that instances in safe areas carry higher weight than those in borderline or overlapping areas. Hence, oversampling is prioritized for instances with higher weights, while undersampling is the opposite. In contrast, OCSV-US prioritizes only the instances in borderline areas and removes the rest. NROMM and GLOS categorize the instances into several groups based on their difficulty. Unlike SOUP and MC-EVHS, no weight is assigned to each instance. Each category has its own resampling method, as it has distinct characteristics that differ from those of the others. NROMM uses interpolation (SMOTE) for instances in inland and borderline areas, while adaptive embedding is used for instances in trapped areas. Meanwhile, GLOS uses MMO for instances in safe areas, AMDO for instances in borderline areas, and RRO for instances considered as small disjunct.

On the other hand, other resampling methods simply use random selection to choose the observed instances, i.e., MC-RBO, MC-CCR, OREM-M, and MC-NRO. Uniform distribution is commonly used. This approach is straightforward but may ignore important instances. Therefore, SHSampler proposes weighted random sampling through a roulette wheel mechanism. Instances with higher weights have a higher probability of being chosen.

In summary, Table 4 presents the advantages and disadvantages of each data resampling method in the context of multiclass imbalance learning.

Table 4 Advantages and disadvantages of several ad hoc data resampling methods for multiclass imbalance learning

Methods	Advantages	Disadvantages
S-SMOTE	Uses a memetic algorithm for optimization. Uses RBFNNs instead of decision trees	Computationally expensive. Ignores class overlap and small disjunct
SCUT	Uses a hybrid resampling. Addresses small disjunct	Does not address class overlap. Potentially increases class overlap
MDO	Synthetic samples reflect the class variation. Avoids generating overlapping synthetic samples	Requires fine-tuned parameters for optimal results. Keeps overlapping samples from the original set
SMOM	Expands the area of minority class regions. Avoids generating overlapping synthetic samples	Computationally expensive. Keeps overlapping samples from the original set
SOUP	Considers the safe level of each sample. Helps reduce overlap between classes	Uses data duplication for oversampling. Minority classes are less exposed
MC-RBO	Expands the area of minority class regions. Addresses small disjunct	Requires fine-tuned parameters for optimal results. Oversampling is performed on randomly selected instances
MC-CCR	Expands the area of minority class regions. Addresses small disjunct	Requires fine-tuned parameters for optimal results. Changes the original samples due to data translation
OCSV-US	Results in a significantly smaller dataset after resampling. Reduces the impact of class overlap	Requires fine-tuned parameters for optimal results. Computationally expensive
SHSampler	Expands the area of minority class regions. Reduces the impact of class overlap and small disjunct	Performs poorly on datasets with extreme data distributions. Computationally expensive
MC-EVHS	Considers the safe level of each sample. Reduces the impact of class overlap and small disjunct	Ignores local information in the dataset. Tested only on a few datasets with moderate difficulty
NROMM	Considers the safe level of each sample. Reduces the impact of class overlap and small disjunct	Performs poorly on datasets with many borderline samples. Computationally expensive
OREM-M	Expands the area of minority class regions. Reduces the impact of class overlap and small disjunct	Oversampling is performed on randomly selected instances. Computationally expensive
GLOS	Uses various resampling strategies tailored to each data difficulty. Generates more diverse synthetic samples	Requires fine-tuned parameters for optimal results. Keeps overlapping samples from the original set
MC-NRO	Reduces the impact of class overlap. Expands the area of minority class regions	Requires fine-tuned parameters for optimal results. Oversampling is performed on randomly selected instances

5.2 Ensemble learning approach

Ensemble learning emerges as an alternative approach to dealing with imbalanced datasets since it can diversify the classifiers and combine their results (Galar et al. 2012). This strategy enables the classifiers within the ensemble mechanism to achieve better generalizations when performing classification (Kumar et al. 2019). Like the data-level approach, ensemble learning is independent of the classification algorithm. Therefore, it can work with any machine learning algorithm. To improve the performance in handling imbalanced datasets, ensemble learning is usually combined with a resampling method, such as RUSBoost (Seiffert et al. 2010), EasyEnsemble (Liu et al. 2009), SMOTEBagging (Wang and Yao 2009), SMOTEBoost (Chawla et al. 2003), UnderBagging (Barandela et al. 2003), and Balanced-Bagging (Opitz and Maclin 1997). However, these methods are designed to work in binary class datasets.

While most ensemble learning methods employ machine learning algorithms as base classifiers, some frameworks instead use fuzzy systems based on the Takagi-Sugeno-Kang (TSK) model as base classifiers (Zhang et al. 2023, 2024). In both frameworks, random undersampling (RUS) is employed to address class imbalance, whereas the TSK fuzzy system is used to compute sample sensitivity. Moreover, ensemble learning frameworks incorporating TSK fuzzy systems represent a distinct research trajectory, separate from those based on traditional machine learning algorithms. This distinction is reflected in the substantial body of literature dedicated to this specific approach. Recent studies on handling class imbalance include those by Wang et al. (2022); Bian et al. (2023); Zhang et al. (2023). Therefore, this review excludes studies related to ensemble learning based on TSK fuzzy systems.

Several ensemble mechanisms have also been proposed to address the multiclass imbalance learning problem specifically. Each method works based on a bagging or boosting mechanism integrated with a data-level approach, specifically data resampling or feature engineering. Tables 5 and 6 list 12 ensemble methods discussed in this paper, filtered from selected highly reputable publications. The following discussion on the ensemble approach in multiclass imbalance learning is presented from several aspects: ensemble mechanism, ensemble algorithm, base classifier, data processing strategy, and aggregation technique.

5.2.1 Ensemble mechanisms

Ensemble learning has three mechanisms: stacking, bagging, and boosting (Mienye and Sun 2022). Other works add voting as one of the mechanisms (Mahajan et al. 2023; Manconi et al. 2022). Each has the same purpose: to improve the classifier's generalization ability in performing classification. However, some differences in their work and basic structures lead to different effectiveness and efficiency in handling imbalanced datasets. Figure 8 illustrates these mechanisms.

Voting, the simplest and most straightforward mechanism, employs n classifiers as base learners and determines the final prediction through a majority vote. It can also conduct voting based on each classifier's weight or competence, known as hard voting and soft voting, respectively. The base learners can be different machine learning algorithms or the same algorithm with different parameter settings, providing a reassuringly simple approach to ensemble learning. On the other hand, stacking can be viewed as an extension of the voting mechanism, having the same structure as voting but a different mechanism for obtaining the final result. Instead of using hard or soft voting, stacking uses a meta-learner to combine all results from the classifiers. The meta-learner can be any machine learning algorithm that learns from the patterns in the results of the base learners. Figure 8 shows that voting and stacking use the whole dataset to train the n classifiers.

In multiclass imbalance learning, voting and stacking are unsuitable for dealing with imbalanced datasets because they lack diversity in the dataset learned by each classifier. However, to build an optimal ensemble classifier, it is necessary to have diversity in the dataset to minimize bias-variance and ambiguity (Galar et al. 2012) and also to increase generalization (Li et al. 2008). Data diversity is a key factor that enables each classifier

Table 5 Summary of ensemble learning approaches in multiclass imbalance learning

Method	Year	Ensemble learning specification					Method's description
		Mechanism	Algorithm	Base classifier	Data processing	Aggregation	
Easy-BPNN	2016	Boosting	AdaBoost	Neural Network	Undersampling	Majority voting	Easy-BPNN uses OvO decomposition to handle multiclass datasets. It utilizes the EasyEnsemble framework, with a backpropagation neural network (BPNN) serving as the base classifier. Random undersampling is employed within each bootstrap in the EasyEnsemble framework to balance the class distribution.
AMCS	2016	Boosting	AdaBoost.M1	SVM, KNN, C4.5, RBFNN, DGC	Feature selection	Majority voting	AMCS employs the AdaBoost algorithm with different base classifiers depending on the type of imbalanced dataset. It also incorporates feature selection using fast correlation-based filter (FCBF) and binary particle swarm optimization (BPSO), depending on the dataset type.
DES-MI	2018	Bagging	Random Balance	CART	Hybrid resampling	Majority voting	DES-MI generates bootstraps of varying sizes with balanced class distributions. It employs dynamic ensemble selection to choose the most competent classifiers based on the neighborhood of the observed test instance.
DRCW-ASEG	2018	Bagging	–	CART	Oversampling	Majority voting	DRCW-ASEG generates bootstraps using standard sampling with replacement. During testing, it finds the nearest neighbors of each test instance and calculates the local imbalance ratio. If the imbalance ratio exceeds a defined threshold, a SMOTE-like approach is applied to balance the neighborhood. Predictions for each bootstrap are made based on the distances to neighboring instances.
PT-Bagging	2018	Bagging	–	J48	-	Weighted voting	PT-Bagging generates bootstraps using standard sampling with replacement. For each test instance, it calculates the probability of belonging to each class and then calculates the average class probabilities from all base classifiers. The class with the highest probability is returned as the final prediction.
EVINCI	2019	Bagging	–	C4.5	Undersampling	Majority voting	EVINCI generates subsets using random sampling with varying imbalance ratios. For each subset, a genetic algorithm is used to filter instances to maximize the accuracy of the ensemble. The ensemble with the highest accuracy is selected as the optimal model and used during the testing phase.

to learn distinct patterns, resulting in classification models that differ from one another. Therefore, bagging and boosting are the preferred ensemble mechanisms for dealing with multiclass imbalanced datasets, as shown in Tables 5 and 6.

Bagging (bootstrap aggregating) creates n subsets from the original dataset. Each subset contains m instances randomly selected from the original dataset using sampling with a replacement procedure. This is known as the bootstrap sampling method. As shown in Fig. 8, the n subsets produce n different classification models aggregated using the voting mechanism to obtain the final prediction. In bagging, each model is independent of each other. Most of the previous works use bagging as the ensemble mechanism, i.e., DES-MI (García et al. 2018), DRCW-ASEG (Zhang et al. 2018), PT-Bagging (Collell et al. 2018), EVINCI (Fernandes and de Carvalho 2019), MultiRandBal (Rodríguez et al. 2020), DPSE (Gao et al. 2021), and E-EVRS (Grina et al. 2023).

In contrast, the boosting mechanism does not employ a sampling strategy to create diversity in the dataset; instead, it assigns weights to each instance to represent the importance of those instances in the learning process. The weight of one instance will be increased in the next iteration if misclassified in the current iteration. To carry out this mechanism, boosting must work sequentially to learn the dataset with different weight compositions iteratively. This contrasts with bagging, which works in parallel. The number of iterations determines the number of classification models (base learner) produced. The final prediction is obtained based on the majority vote of the base learners. Several previous works use boosting mechanisms, as in Easy-BPNN (Zhang et al. 2016), AMCS (Yijing et al. 2016), MultiRandBal (Rodríguez et al. 2020), OREMBoost (Zhu et al. 2023), and DOMIN (Jedrzejowicz and Jedrzejowicz 2023).

According to Juez-Gil et al. (2021), boosting outperforms bagging. The sequential learning mechanism in boosting has a significant impact on producing excellent generalization ability (Tanha et al. 2020). However, to achieve optimal performance, several key parameters must be carefully determined. In addition, since sequentially boosting requires longer computational time than bagging, which can work in parallel, it is less efficient. It is worth noting that bagging is also a powerful and straightforward method for addressing the imbalance problem, as it only requires a few parameters to be set (Galar et al. 2012).

5.2.2 Ensemble algorithms

It is worth distinguishing between ensemble mechanism and algorithm when working on ensemble learning. In this paper, ensemble mechanisms refer to voting, stacking, bagging, or boosting, representing how the ensemble generally works. Ensemble algorithms, on the other hand, represent the more specific learning procedure within the ensemble mechanism. For example, in the boosting mechanism, several learning algorithms are employed, including AdaBoost (Freund and Schapire 1997), GradientBoost (Friedman 2001), and XGBoost (Chen and Guestrin 2016). Meanwhile, in bagging, a random balance algorithm (Díez-Pastor et al. 2015) is employed during the bootstrapping process.

All previous works in Tables 5 and 6 that utilize the boosting mechanism employ AdaBoost (adaptive boosting) as the ensemble algorithm. With its simple yet effective approach, this algorithm works by adaptively adjusting the weight of the instance to correct errors from the previous learning iteration. It is proposed using a weak learner called a decision stump, a one-level decision tree containing only a root and two leaves. However, other

Table 6 Summary of ensemble learning approaches in multiclass imbalance learning (continued from Table 5)

Method	Year	Ensemble learning specification					Method's description
		Mechanism	Algorithm	Base classifier	Data processing	Aggregation	
MultiRandBal	2020	Bagging/ Boosting	Random Balance	Decision Tree	Hybrid resampling	Majority voting	MultiRandBal generates bootstraps with a random portion assigned to each class. For each bootstrap, SMOTE is applied to any class whose size is smaller than the assigned portion. The final prediction is obtained through a simple majority voting mechanism.
DPSE	2021	Bagging	Different Partition Sampling	CART, LR, KNN, SVM, MLP, RF	Hybrid resampling	Weighted voting	DPSE uses OvO decomposition and generates bootstraps with gradually increasing sizes. For each OvO subset, BorderlineSMOTE is applied to augment the minority class, while s-Random Undersampling reduces the majority class to match the target size. Final predictions are aggregated using weighted voting based on class confidence.
OREMBoost	2023	Boosting	AdaBoost	C4.5, SVM, Neural Network	Oversampling	Majority voting	OREMBoost is an AdaBoost-based algorithm that incorporates OREM-M resampling to balance class distribution in each boosting iteration.
E-EVRS	2023	Bagging	-	Decision Tree, SVM	Hybrid resampling	Weighted voting	E-EVRS is a bagging-based method that incorporates MC-EVHS resampling in each bootstrap to balance class distribution. The final prediction is obtained through a weighted voting mechanism, which uses Dempster's rule to combine class probabilities from each base classifier.
DOMIN	2023	Boosting	AdaBoost	Gene Expression Programming	Hybrid resampling	Majority voting	DOMIN is an AdaBoost-based model that employs OvO decomposition to handle multiclass datasets, using gene expression programming as the base classifier. In each boosting iteration, dominance-based oversampling, which is based on genetic algorithm selection, is used to filter the instances.
AdaBoost.AD	2024	Boosting	AdaBoost with adaptive weights	Decision Tree	–	Majority voting	AdaBoost.AD is an AdaBoost-based model extended to incorporate the density information of training instances into the class imbalance ratio, allowing instances from the same class to be assigned different weights. These weights are used to determine the weight of each base classifier and are then updated in the next iteration.

machine learning algorithms can also be used. Despite its simplicity, it can effectively deal with imbalanced datasets when combined with a data resampling approach, such as SMOTE (Tanha et al. 2020).

Recent work by Li et al. (2024) proposed AdaBoost.AD, which extends AdaBoost by using adaptive sample weighting to replace the conventional weighting in the standard AdaBoost algorithm. Instead of calculating the weight of instances based on the entire training set, AdaBoost.AD takes into account the density information around each sample to determine its weight. Thus, samples from the same class may have different weights.

Other previous works use the bagging mechanism. Most bagging-based methods do not have any specific algorithm because bagging is a simpler mechanism than boosting. The primary process in bagging is bootstrapping, which creates several subsets. Previous works, such as MultiRandBal and DES-MI, employ random balance algorithms to enhance this process. Random balance is a technique that creates n subsets from the original dataset where the ratio between classes is randomized for each subset.

In Fig. 9a, DES-MI generates bootstraps of varying sizes with balanced class distributions. In some cases, the bootstrap size can exceed that of the original dataset. On the other hand, DPSE enhances the DES-MI bootstrapping strategy through an algorithm called differential partial sampling. As shown in Fig. 9b, DPSE employs a one-vs-one (OvO) mechanism, where each subset contains only two classes and gradually increases in size at each iteration. Meanwhile, MultiRandBal applies a random balance algorithm to generate bootstraps with the same total size but varying class distributions, as illustrated in Fig. 9e. In all these three frameworks, SMOTE is employed to augment classes whose sizes are smaller than the defined ratio.

EVINCI uses a multi-objective evolutionary algorithm (MOEA) based on the NSGA-II algorithm to optimize sample selection for generating bootstraps. Based on Fig. 9c, each class in EVINCI always has a size equal to or smaller than that in the original dataset, as it does not perform oversampling in any of the bootstraps. However, this strategy is computationally expensive due to the use of the evolutionary algorithm. E-EVRS generates bootstraps with equal class sizes, as shown in Fig. 9f. It calculates the average class size from the dataset and performs oversampling on classes whose sizes are smaller than the average class size. Otherwise, undersampling is employed. While other frameworks employ specific algorithms, PT-Bagging utilizes sampling with replacement, similar to standard bootstrapping. As a result, it produces subsets of equal size containing randomly distributed class instances, as illustrated in Fig. 9d.

5.2.3 Base classifiers

Tables 5 and 6 indicate that various machine-learning algorithms have been employed in previous studies. The most prevalent algorithms are decision tree-based classifiers, such as C4.5, J48, and CART. While the impact of the classification algorithm on the quality of the resulting model is not as significant as the influence of ensemble mechanisms, it remains a critical factor. Different algorithms possess distinct strengths and limitations, which can affect their performance on particular datasets.

Decision tree-based classifiers are often preferred due to their interpretability and ability to effectively handle both numerical and categorical data. Furthermore, the structure of decision trees is particularly adept at capturing complex decision boundaries, which is

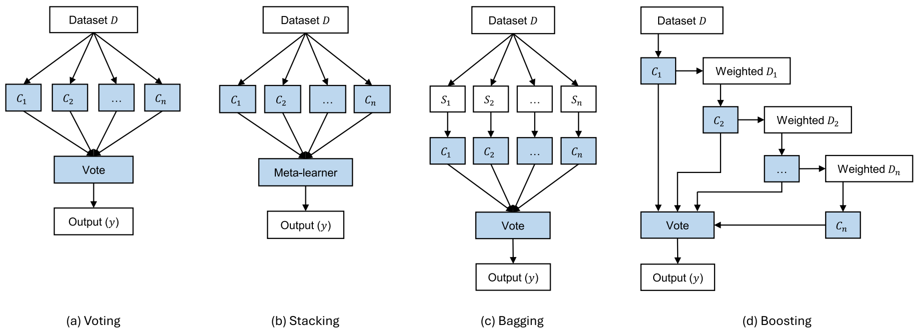


Fig. 8 Various ensemble mechanisms using n classifiers $(C_1, \ldots, C_n)$. Bagging generates n subsets $(S_1, \ldots, S_n)$ through bootstrap sampling, whereas boosting has n different weighted datasets (Weighted $D_1, \ldots$,Weighted D_n)

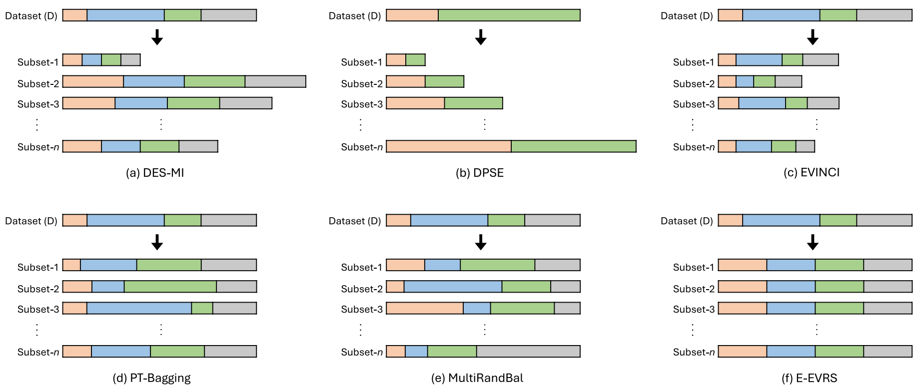


Fig. 9 Illustration of different bootstrap strategies in various ensemble frameworks using the bagging mechanism. Classes in the dataset are represented in different colors and sizes

essential for achieving high accuracy in classification tasks. Therefore, carefully selecting an appropriate algorithm that matches the dataset's characteristics and the problem's specific requirements is crucial for optimizing the performance of the classification model.

On Easy-BPNN, using neural networks with a backpropagation algorithm as base classifiers effectively outperforms ensemble learning with a decision tree as the base classifier. The non-linearity of neural networks helps them classify complex data distributions. However, neural networks require appropriate architecture and parameter settings for optimal performance. Additionally, the learning process takes significantly longer than that of decision tree-based classifiers, especially when working with large datasets.

In addition, another neural network architecture, namely RBFNN, is known to be effective in addressing class imbalance issues (Pérez-Godoy et al. 2014; Ganapathy et al. 2015). It is also relatively faster to train than neural networks with the backpropagation algorithm (BPNN). Several works integrate RBFNN with cost-sensitive learning to enhance the classification performance on imbalanced datasets (Kumudha and Venkatesan 2016; Wu et al. 2020). However, this method is rarely used in recent studies as an ensemble-based classifier, particularly in multiclass imbalance problems. AMCS is the only ensemble framework, and

S-SMOTE is the only data resampling approach that uses RBFNN as one of the base classifiers. We believe that it is worth evaluating the performance of RBFNN as a base classifier in ensemble learning frameworks in future work.

5.2.4 Data processing strategies

Integrating data resampling into ensemble learning can enhance the learning process on imbalanced datasets, as many recent works that use this strategy have proven (Kumar et al. 2019). Werner de Vargas et al. (2023) asserted that classification tasks using ensemble learning on an imbalanced dataset perform better when data processing is applied. Two main data processing approaches have been used in previous works, as listed in Tables 5 and 6.

5.2.4.1 Data resampling approach The first approach involves data resampling of the imbalanced dataset, i.e., oversampling, undersampling, or hybrid resampling. This approach can be applied through two strategies, as depicted in Fig. 10. The first strategy involves applying data resampling to the entire dataset (outside the ensemble mechanism) before using it as input to ensemble learning. The second approach consists of applying data resampling within the ensemble mechanism before training each classifier. In the bagging mechanism, the second strategy is applied to each subset generated from bootstrap sampling. On the other hand, it is applied to each weighted dataset in every iteration of the boosting mechanism. Juez-Gil et al. (2021) stated that the second strategy is better than the first one. It is worth noting that when working with big data, integrating ensemble learning with oversampling methods may be ineffective, as they are typically designed for normal-sized datasets (Juez-Gil et al. 2021).

Most methods in Tables 5 and 6 use data resampling to improve the ensemble learning performance. Easy-BPNN, DES-MI, MultiRandBal, DPSE, OREMBoost, and E-EVRS utilize data resampling within their ensemble mechanisms. Meanwhile, DOMIN implements data resampling outside the ensemble mechanism. In general, data resampling is used in the training process of classifiers. However, DRCW-ASEG (Zhang et al. 2018) uses it in the testing or inference phase using the ASEG algorithm. It first identifies k nearest neighbors from the testing data and then generates synthetic instances within the neighborhood if the prior training set has an imbalance ratio below a certain threshold. On the other hand,

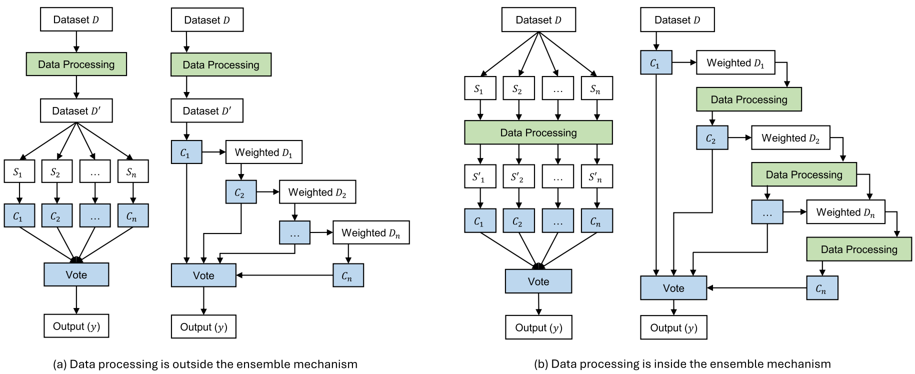


Fig. 10 Different scenarios for integrating data processing into bagging and boosting

AMCS uses feature selection by combining a fast correlation-based filter and binary particle swarm optimization.

5.2.4.2 Feature engineering approach The second approach to data processing involves conducting feature engineering, which includes feature extraction and feature selection. Feature extraction generates features from the raw dataset or manipulates the existing features to generate new derivative features. These can be accomplished using a statistical approach, such as calculating the minimum, maximum, median, mean, standard deviation, and variance, or by employing specific algorithms, like PCA and t-distributed stochastic neighbor embedding, to transform the existing features into a new feature space. In contrast, instead of generating new features, feature selection reduces the number of features and retains the most significant ones that impact classification. It can be performed using statistical analysis, such as variance threshold, information gain, and feature correlation, or using evolutionary computation, such as a genetic algorithm. Similar to the data resampling approach, feature engineering can be applied either outside or inside the ensemble mechanism. However, feature engineering is not used as frequently as data resampling, as indicated by previous works.

Recent approaches in feature engineering leverage multi-view learning, which integrates features from multiple views or sources originating from diverse domains, thus enhancing generalization capabilities (Yu et al. 2024). A straightforward example of this approach is multi-view speech recognition, which combines audiovisual features from a speaker's utterances and lip movements. In the context of class imbalance, Tang et al. (2023) incorporated multi-view learning with cost-sensitive learning to address class imbalance in classification tasks. Another study by Graa and Rekik (2019) integrated multi-view learning with manifold learning and SMOTE-based oversampling. It uses PCA with multiple Gaussian kernels to generate multiple views as inputs to the classifier, where SMOTE is employed to balance the class distribution within each view. We believe using multi-view learning could be beneficial in future work to develop more effective approaches for multiclass imbalance learning.

5.2.5 Aggregation methods

In ensemble learning, aggregation is a mechanism that combines all the classifier predictions to obtain the final prediction, typically performed using the majority vote strategy. The prediction with the most votes will be assigned as the final result. Several ensemble frameworks propose specific mechanisms to achieve better majority voting results, as in DES-MI and DRCW-ASEG. DES-MI employs a dynamic ensemble selection (DES) strategy to use only the predictions from the most competent classifiers. This approach aims to ensure that the final predicted class is based on high-confidence predictions from the selected classifiers. Meanwhile, DRCW-ASEG conducts a specific data resampling method, Adaptive Synthetic Example Generation (ASEG), in the testing phase to balance the class distribution in a local neighborhood around it. This strategy enables the framework to operate dynamically and adaptively, depending on the test data it receives.

Another strategy is using soft or weighted voting, as in PT-Bagging, E-EVRS, and DPSE. PT-Bagging takes into account the predicted probabilities from each classifier and calculates their average for each class. The class with the highest average probability is chosen as the final prediction. Similar to PT-Bagging, E-EVRS also uses probability from the predicted class in each classifier as the basis to predict the final result through Dempster's rule. On the other hand, DPSE takes the confidence values from all base classifier's predictions and calculates the average to obtain the final prediction. Generally, a weighted vote can be more accurate than a majority vote since it considers the competence of each base classifier.

In summary, Table 7 presents the advantages and disadvantages of each ensemble learning framework in the context of multiclass imbalance learning.

Table 7 Advantages and disadvantages of several ensemble learning frameworks in multiclass imbalance learning

Methods	Advantages	Disadvantages
Easy-BPNN	Uses neural networks as base classifiers. Incorporates bagging and boosting mechanisms	High computational cost due to the use of neural networks. Does not address small disjunct
AMCS	Incorporates data resampling and feature selection. Uses adaptive multiple classifier systems	Potentially increases class overlap due to the use of SMOTE. Data resampling is not optimally used in AMCS
DES-MI	Uses dynamic selection in the voting mechanism. Enhances diversity in the base classifier	Requires fine-tuned parameters for optimal results. Requires high computation in the voting mechanism
DRCW-ASEG	Adaptive, as resampling is based on the testing sample. Has an efficient training phase	Requires high computation during the testing phase. Requires fine-tuned parameters for optimal results
PT-Bagging	Adaptive, as it uses threshold-moving in the voting phase. Has an efficient training phase	Has less diverse classifiers than other frameworks. Performance on the GM metric has not been evaluated
EVINCI	Generates a smaller size of bootstrap than the original set. Ables to select the significant sample into the bootstrap	Computationally expensive due to the use of NSGA-II. Requires fine-tuned parameters for optimal results
MultiRandBal	Flexible to use bagging or boosting mechanisms. Generates diverse classifiers on the boosting mechanism	Potentially increases class overlap due to the use of SMOTE. Computationally expensive when using a boosting mechanism
DPSE	Enhances diversity in the base classifier. Uses sample safe-level to reduce class overlap	Computationally expensive due to the OvA approach. Performance on the GM metric has not been evaluated
OREMBoost	Expands the area of minority class regions. Reduces the impact of class overlap and small disjunct	Oversampling is performed on randomly selected instances. Computationally expensive
E-EVRS	Reduces the impact of class overlap and noise. Enhances diversity in the base classifier	Requires fine-tuned parameters for optimal results. Potential bias in the centroid calculation
DOMIN	Optimizes the oversampling using genetic algorithms. Reduces the impact of class overlap	Computationally expensive due to the OvO approach. Requires fine-tuned parameters for optimal results
AdaBoost.AD	Uses adaptive weights for each sample. Incorporates the density information of the training set	Has low performance on multi-majority datasets. Not evaluated on datasets with very small classes

6 Datasets on multiclass imbalance learning

The characteristics of a multiclass imbalanced dataset differ from those of a binary one. Wang and Yao (2012) distinguished two types of multiclass imbalanced datasets, i.e., multi-majority and multi-minority, which refer to datasets with many majority classes and only one minority class and vice versa. An additional type, linear imbalance, introduced by Buda et al. (2018), is a dataset with a linearly increasing class size. More generally, it can also be called gradual imbalance (Lango and Stefanowski 2022).

It is easier to determine whether a dataset is multi-majority or multi-minority than to determine if it has a gradual imbalance. Therefore, this paper classifies the dataset into either multi-majority or multi-minority categories in advance and then determines the rest as a gradual imbalance. To establish if a class is majority or minority, a certain threshold is required. Fernandes and de Carvalho (2019) proposed a method to calculate the threshold based on the average and standard deviation of all class sizes. However, this approach can produce biased thresholds due to the influence of the central tendency of the class size distribution. Instead, this study decides to define the threshold as the average of the largest and smallest classes:

$$\text{threshold} = \frac{N_{\max} + N_{\min}}{2}, \tag{1}$$

where $N_{\max}$ and $N_{\min}$ are the amount of instances in the largest and smallest classes, respectively. Equation (1) does not depend on the overall distribution of all classes to avoid central tendencies caused by the dominance of the majority or minority class in the dataset.

Furthermore, this paper underscores the role of the 5-nearest neighbors analysis, a method recommended by Lango and Stefanowski (2022) and influenced by the work of Napierala and Stefanowski (2016), in identifying overlapping instances within the datasets. This method is instrumental as it identifies instances with more neighbors from other classes within its five nearest neighbors, thereby categorizing them as overlapping instances. The percentage of class overlap is further categorized into three class interactions: majority-majority, majority-minority, and minority-minority. Notably, multi-minority datasets have zero overlapping instances in majority-majority interaction due to their single minority class, while multi-majority datasets have zero overlapping instances in minority-minority interaction.

Table 8 provides a detailed specification of 19 widely used datasets in multiclass imbalance learning research, which vary in size and number of classes. Thyroid is the largest, followed by Page Blocks with 7,200 and 5,472 instances, respectively. Yeast and Led7Digit have 10 classes, the most classes among the datasets. Meanwhile, Shuttle is the most imbalanced dataset, having the highest imbalance ratio of 853.0. In contrast, Wine has the lowest imbalance ratio, approximately 1.5. Although the number of classes and imbalance ratio may have a negative impact on classification, the percentage of class overlap still plays a more significant role in degrading classification performance.

The multi-majority datasets have the highest overlap rate among all types of imbalances. The contraceptive dataset proves to be the most challenging, with a 92.60% overlap rate, the highest among the others. It is followed by Cleveland and Winequality-red, with an overlap percentage of 91.42% and 90.24%, respectively. According to the overlap percent-

Table 8 Detailed specification of datasets widely used in multiclass imbalance learning research. IR is the imbalance ratio between the largest and the smallest class. Average accuracy (AvAcc) is the average recall from all classes after training and testing using C4.5 through 5 times 10-fold cross-validations. The percentage of class overlap represents the overlap rate between different class interactions (MajMaj, MajMin, and MinMin)

Dataset	Class	Data	IR	Class distribution	AvAcc	% of class overlap			
					(%)	MajMaj	Maj-Min	Min-Min	Overall
Gradual imbalance									
Winequality-red	6	1599	68.10	681/638/199/53/18/10	36.18	40.40	49.84	0.00	90.24
Yeast	10	1484	92.60	463/429/244/163/51/44/35/30/20/5	44.49	49.39	33.42	5.93	88.74
Glass	6	214	8.44	76/70/29/17/13/9	67.98	23.83	42.06	1.87	67.76
Ecoli	8	336	71.50	143/77/52/35/20/5/2/2	62.92	3.87	36.01	5.95	45.83
Dermatology	6	358	5.55	111/71/60/48/48/20	92.67	0.00	11.20	28.96	40.16
Flare	6	1066	7.70	331/239/211/147/95/43	59.88	21.48	35.73	1.13	58.34
Multi-majority									
Contraceptive	3	1473	1.89	629/511/333	45.94	25.05	67.55	0.00	92.60
Balance-scale	3	625	5.88	288/288/49	56.51	9.76	31.68	0.00	41.44
Led7digit	10	500	1.54	57/57/53/52/52/51/49/47/45/37	68.59	52.60	24.20	0.00	76.80
Lymphography	4	148	40.5	81/61/4/2	67.12	60.14	6.08	0.00	66.22
UKM	4	403	2.6	129/122/102/50	90.82	34.74	25.31	0.00	60.05
Hayes-Roth	3	160	2.10	65/64/31	84.29	65.00	26.25	0.00	91.25
Multi-minority									
Cleveland	5	297	12.62	164/55/36/35/13	29.38	0.00	85.15	6.27	91.42
Newthyroid	3	215	5.00	150/35/30	92.51	0.00	19.53	0.00	19.53
Page Blocks	5	5472	175.46	4913/329/115/87/28	83.29	0.00	9.96	0.51	10.47
Wine	3	178	1.48	71/59/48	92.77	0.00	57.87	3.37	61.24
Thyroid	3	7200	40.16	6666/368/166	98.30	0.00	19.49	0.31	19.80
Zoo	7	101	10.25	41/20/13/10/8/5/4	99.33	0.00	1.98	16.83	18.81
Shuttle	5	2175	853.00	1706/338/123/6/2	87.74	0.00	1.75	0.23	1.98

age, these three datasets have a low average accuracy of 45.94%, 29.38%, and 36.18%, respectively. Contraceptive and Cleveland have the highest overlap between majority and minority classes (MajMin).

On the other hand, the multi-minority datasets have the lowest overlap rate, excluding Cleveland. Therefore, the average accuracies of this type of dataset are mostly higher than those of gradual imbalance and multi-majority. This paper considers Cleveland multi-minority instead of gradual imbalance, as Lango and Stefanowski (2022) did, because the distribution of classes in Cleveland does not show a gradual increase in class size but rather tends to be a multi-minority class distribution.

The average accuracy presented in Table 8 is a crucial metric influenced by the overlap between the majority classes (MajMaj) and the overlap between the majority and minority classes (MajMin). The trend is clear: as the overlap rate in MajMaj or MajMin increases, the average accuracy tends to decrease.

7 Evaluation metrics in multiclass imbalance learning

When conducting a classification task, accuracy is the most commonly used metric to measure the performance of the proposed model. It simply calculates the percentage of correct predictions from all classes. However, this metric is unsuitable for classification with imbalanced datasets, as the majority class often dominates correct predictions. Therefore, alternative metrics are suggested to evaluate the model's balanced performance across classes. Based on recent studies, several evaluation metrics are commonly used in multiclass imbalance learning. Let m be the number of classes; the following specify the most common evaluation metrics in multiclass imbalance learning:

1. Geometric mean (GM). In binary classification, this metric calculates the square root of the product of the recall values from both classes. This metric can be extended into multiclass GM (mGM) by changing the square root into the m-th root, namely (Krawczyk et al. 2020; Koziarski et al. 2020):

$$\mathrm{mGM} = \left(\prod_{i=1}^{m} \mathrm{recall}_i\right)^{\frac{1}{m}}. \tag{2}$$

   This metric is sensitive to class imbalance, as it will yield a low score if one class has a low recall rate. In extreme conditions, where one class has a recall of 0, the overall mGM will be 0.
2. Average accuracy (AvAcc). This metric simply calculates the average of recalls from all classes in the dataset, namely (Krawczyk et al. 2020; Koziarski et al. 2020):

$$\mathrm{AvAcc} = \frac{1}{m}\sum_{i=1}^{m} \mathrm{recall}_i. \tag{3}$$

   Like mGM, AvAcc is also sensitive to class imbalance; however, this metric still yields a more moderate result than GM, even when one class has a recall of 0.
3. Area under ROC curve (AUC). This metric calculates the area under the receiver operating characteristic (ROC) curve, which is created by plotting the true positive rate (TPR) and false positive rate (FPR) at different decision thresholds. AUC is defined by (Abdi and Hashemi 2016; Yang et al. 2018; Tanha et al. 2020):

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(\mathrm{FPR})\, d(\mathrm{FPR}), \tag{4}$$

   where:

$$\text{TPR} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}, \tag{5}$$

$$\text{FPR} = \frac{\text{False Positive}}{\text{False Positive} + \text{True Negative}}. \tag{6}$$

When working with multiclass datasets, AUC can be extended into multiclass AUC (MAUC) by averaging AUC scores across all pairs of classes, namely:

$$\text{MAUC} = \frac{2}{m(m-1)} \sum_{i=1}^{m} \sum_{j=1}^{m} \text{AUC}_{ij}, \tag{7}$$

where AUC_{ij} is the area under the curve between class *i* and class *j*.

4. F1-score. This metric calculates the harmonic mean from precision and recall, providing an alternative perspective to GM and AvAcc, which only use recall as the basis of measurement. For binary classification, the F1-score is defined by Collell et al. (2018); Tanha et al. (2020):

$$\text{F1} = 2\frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}. \tag{8}$$

In multiclass classification, the macro F1-score is calculated by averaging the F1-score from all classes, namely:

$$\text{MacroF1} = \frac{1}{m} \sum_{i=1}^{m} \text{F1}_i, \tag{9}$$

where F1_i is the F1-score of class *i*.

5. Class-balance accuracy (CBA). This metric takes into account the imbalance between the predictions and the actual class distribution. CBA is defined by (Krawczyk et al. 2020; Koziarski et al. 2020):

$$\text{CBA} = \frac{1}{m} \sum_{i=1}^{m} \frac{\text{mat}_{i,i}}{\max\left(\sum_{j=1}^{m} \text{mat}_{i,j}, \sum_{j=1}^{m} \text{mat}_{j,i}\right)}, \tag{10}$$

where $\text{mat}_{i,j}$ represents the number of instances from class *i* predicted as belonging to class *j*.

6. Confusion entropy (CEN). This metric adopts the concept of information theory, using entropy to measure the consistency (or uniformity) of a classification model's performance. CEN is defined by (Krawczyk et al. 2020; Koziarski et al. 2020):

$$\text{CEN} = \sum_{i=1}^{m} \text{P}_i \text{CEN}_i, \tag{11}$$

where:

$$\mathrm{P}_i = \frac{\sum_{j=1}^{m}(\mathrm{mat}_{i,j} + \mathrm{mat}_{j,i})}{2\sum_{j=1}^{m}\sum_{k=1}^{m}\mathrm{mat}_{j,k}}, \tag{12}$$

$$\mathrm{CEN}_i = -\sum_{j=1, j\neq i}^{m}\left(P_{i,j}\log_{2(m-1)}(P_{i,j}) + P_{j,i}\log_{2(m-1)}(P_{j,i})\right), \tag{13}$$

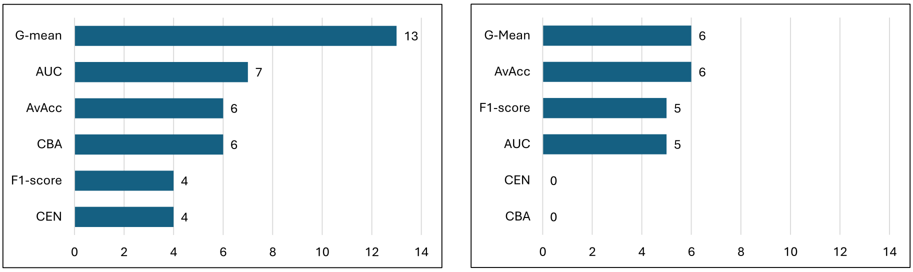
$$\mathrm{P}_{i,j} = \frac{\mathrm{mat}_{i,j}}{\sum_{j=1}^{m}(\mathrm{mat}_{i,j} + \mathrm{mat}_{j,i})}, \quad \text{for } i \neq j, \tag{14}$$

$\mathrm{P}_{i,i} = 0$, and $\mathrm{mat}_{i,j}$ represents the number of instances of the class *i* predicted as class *j*. Since CEN is based on entropy, a lower CEN indicates better model performance, suggesting that the model tends to perform consistently.

GM is the most popular metric in recent studies for both data resampling and ensemble learning, as shown in Fig. 11. Meanwhile, AvAcc, AUC, and F1-score are also widely used. Other metrics, such as CBA and CEN, appear in several studies on data resampling methods. Table 9 summarizes the advantages and disadvantages of evaluation metrics for multiclass imbalance learning.

From our perspective, using GM, AvAcc, and F1-score is sufficient to evaluate classification performance and obtain valuable insights comprehensively. GM can reveal if there is a class with significantly lower accuracy than others, indicating an imbalance in performance across classes. Meanwhile, AvAcc provides insight into the central tendency of the model's performance, showing whether the model tends to be effective overall. F1-score accommodates both precision and recall to evaluate classification performance; therefore, it can complement the evaluation provided by GM and AvAcc.

Furthermore, AUC provides performance information derived from class pairs, thereby illustrating how well the model distinguishes between each pair of classes in multiclass scenarios. However, AUC is quite difficult to interpret in multiclass classification as it accommodates many classes. On the other hand, CBA and CEN are more difficult to interpret, as their calculations are not as straightforward as those of GM and AvAcc. Nevertheless, CBA and CEN can still serve as additional evaluation metrics alongside the AUC.



(a) Evaluation metrics used in data resampling methods

(b) Evaluation metrics used in ensemble learning frameworks

Fig. 11 Distribution of evaluation metrics for multiclass imbalance learning used in **a** data resampling and **b** ensemble learning

Table 9 Advantages and disadvantages of several evaluation metrics in multiclass imbalance learning

Metrics	Advantages	Disadvantages
GM	Sensitive to low class performance. Easy to interpret	Does not consider precision
AvAcc	Can capture the central tendency. Easy to interpret	Does not consider precision
AUC	Evaluates all class pairs. Easy to interpret in binary classification	Rather difficult to interpret in multiclass classification
F1-score	Accommodates precision and recall. Sensitive to low class performance	A few misclassified instances can significantly impact the F1-score
CBA	Takes into account the imbalance between actual and predicted distributions	Difficult to interpret
CEN	Measures prediction consistency using entropy theory	Complex to calculate

8 Effectiveness of the state-of-the-art methods

One of the primary objectives of this paper is to assess the effectiveness of various ad hoc methods in multiclass imbalance learning, which is achieved by comparing the performance of each ad hoc method against the baseline performance across multiple datasets. The baseline performance is derived from the classification results obtained without applying imbalance handling techniques. The analysis is conducted in two parts. The first discusses the effectiveness of the data resampling approach using the mGM metric, and the second focuses on ensemble learning analysis based on mGM and AvAcc metrics. Both metrics are based on the experimental results presented in the published articles. It is important to note that not all previous works in Tables 2, 3, 5, and 6 provide the complete experimental results for each dataset. Some of them only summarize the results using statistical calculations or illustrations in graphic form. Besides, in some works, not all datasets listed in Table 8 were used in their experiments.

For each analysis of data resampling and ensemble learning, the effectiveness is measured by comparing the evaluation metrics of the methods. Afterward, the performance of each method is compared with the baseline scores of each dataset to investigate whether the corresponding method makes any improvement. Since the previous works do not provide comparisons with any baseline score, this study performs several classification tasks using the C4.5 algorithm through 10-fold cross-validations to establish the baseline mGM scores and using 5-fold cross-validations to obtain the baseline AvAcc scores. Each classification on a single dataset is repeated five times to get comprehensive results. C4.5 with the 10-fold and 5-fold cross-validations are selected to adapt to the methods used by previous studies in data resampling and ensemble learning, respectively. It is worth noting that the scores of the methods are equally comparable because most works directly use scores from earlier works for comparison, and the rest use similar experimental scenarios.

8.1 Effectiveness of data resampling approach

Of the 14 data resampling methods in Tables 2 and 3, only nine can be compared equally, as presented in Table 10. The latter methods are generally better than the earlier methods. In this paper, the effectiveness of the resampling methods is discussed in two parts: a comparison between methods and a comparison with baseline performance. The first aims to determine which method is the best among all the methods, and the second assesses the improvements each method produces compared to the baseline performance.

8.1.1 Comparison between methods

MC-NRO and NROMM perform better than other resampling methods, as indicated by the dominance of darker colors in Table 10. Although neither method always yields the highest mGM score, they generally achieve higher mGM scores than most other methods. However, MC-NRO performs significantly better than NROMM on datasets with gradual imbalance. This is because MC-NRO effectively combines the oversampling mechanism of MC-RBO with the data translation approach of MC-CCR while also removing majority class instances that overlap with other classes. This strategy enables MC-NRO to generate synthetic instances and expand the area of the minority class while simultaneously reducing the impact of class overlap within the dataset.

Additionally, MC-NRO employs a Gaussian distribution when generating synthetic samples, which results in the synthetic instances clustering near the oversampled minority instances. This allows MC-NRO to expand the minority class regions in a controlled manner, helping to prevent the generation of synthetic samples that overlap with other classes.

In contrast, NROMM outperforms MC-NRO on datasets with multi-majority and multi-minority class configurations. NROMM successfully enhances SMOTE's performance by integrating it with a complex clustering algorithm and a data categorization strategy to prioritize the resampling order. On datasets with a multi-majority configuration, the data trans-

Table 10 Comparison of data resampling performance based on the mGM metric. A darker color indicates a higher score

Dataset	S-SMOTE	MDO	SMOM	SOUP	MC-RBO	MC-CCR	OCSV-US	NROMM	MC-NRO
Gradual Imbalance									
Winequality-red	0.367	0.407	0.426	0.471	0.483	0.467	0.411	0.479	**0.489**
Yeast	0.524	0.511	0.529	0.451	0.581	0.569	NA	0.559	**0.591**
Glass	0.626	0.680	0.680	0.667	0.731	0.711	NA	0.797	**0.807**
Ecoli	0.675	0.710	0.635	0.735	0.727	0.710	NA	0.717	**0.738**
Dermatology	0.952	0.921	0.921	0.962	0.957	0.949	0.913	0.964	**0.967**
Flare	0.619	0.586	0.604	0.566	0.655	0.647	0.624	0.640	**0.663**
Multi-majority									
Contraceptive	0.434	0.517	0.526	**0.535**	0.510	0.497	0.481	0.528	0.529
Balance-scale	0.524	0.559	0.587	0.585	0.619	0.622	**0.867**	0.667	0.644
Led7digit	0.708	0.782	0.773	0.778	0.750	**0.804**	0.707	0.786	0.797
Lymphography	0.583	0.731	0.740	NA	0.791	0.774	NA	**0.882**	0.843
Hayes-Roth	0.846	0.832	0.868	0.835	0.819	0.852	0.789	**0.912**	0.881
Multi-minority									
Cleveland	0.246	0.265	0.246	0.303	0.340	0.310	0.320	**0.360**	0.345
Newthyroid	0.865	0.926	0.935	0.908	0.967	0.922	0.904	0.977	**0.980**
Page Blocks	0.744	0.791	0.772	**0.884**	0.873	0.826	0.784	0.882	**0.884**
Wine	0.934	0.901	0.891	0.910	0.917	0.929	**0.984**	0.971	0.951
Thyroid	0.819	0.759	0.777	0.922	0.869	0.815	**0.981**	0.961	0.931
Zoo	0.635	0.758	0.777	NA	0.850	0.843	NA	**0.969**	0.937

lation and data removal mechanisms in MC-NRO are ineffective due to the extremely high degree of class overlap. In fact, the level of overlap in this configuration is even higher than that observed in gradually imbalanced datasets. This issue is exacerbated by the dominance of majority classes, which leads to the excessive removal and translation of instances from these classes.

Furthermore, the oversampling mechanism in MC-NRO, which is based on MC-RBO, is less effective when applied to multi-minority datasets. Although this technique can expand the minority class regions, oversampling is performed on randomly selected instances from the minority class. As a result, the synthetic instances may be generated in suboptimal areas of the feature space. In contrast, NROMM is highly selective, performing oversampling only on minority instances located in safe or borderline areas. This ensures that the synthetic instances are generated in appropriate regions, thereby increasing the representation and presence of minority instances in the dataset.

MC-RBO and MC-CCR show competitive performance to each other. Both methods have better performance. Using local boundary sampling in both methods is effective in dealing with dataset imbalances while minimizing class overlap. MC-RBO tends to outperform MC-CCR in gradual imbalance and multi-minority datasets. The mechanism for generating synthetic instances based on Gaussian RBF in MC-RBO is a more effective strategy than using a spherical area with a fixed radius, as in MC-CCR. Like MC-NRO, the Gaussian RBF in MC-RBO increases the coverage of minority classes without over-expansion, thus not significantly altering the data distribution in the feature space. Moreover, MC-RBO is more efficient than MC-CCR and several other methods because it only uses oversampling rather than hybrid resampling.

On the other hand, MC-CCR significantly outperforms MC-RBO and other methods only on the Led7Digit dataset, a moderately difficult dataset with an overlap rate exceeding 70%. MC-CCR tends to cause over-expansion due to its use of radius-based oversampling, as the synthetic instances may be generated near the outermost radius, far from the center of the oversampled instances. Furthermore, excessive data translation in MC-CCR risks significantly altering the data distribution and may even increase class overlap.

SOUP, the simplest method, performs well in several datasets. It outperforms earlier methods, which employ more sophisticated mechanisms, such as S-SMOTE, MDO, and SMOM. For the most difficult dataset, Contraceptive, which has the highest overlap rate, SOUP achieves the highest mGM score. The data duplication combined with safe-level mechanisms helps the algorithm avoid generating noise that can increase the overlap rate. Priority-based resampling is highly effective in handling datasets with a high percentage of class overlap.

OCSV-US is the only method that uses evolutionary-based undersampling without any oversampling mechanism. Unfortunately, five datasets are absent from the comparison. Nevertheless, OCSV-US is significantly better than MC-NRO and NROMM for balance-scale datasets, with an improvement of up to 20%. It also achieves the highest mGM scores for Wine and Thyroid datasets. The strategy of maintaining the important support vectors from each class is effective in generating a small yet optimal dataset. However, it may achieve better results if combined with an oversampling process.

The earlier methods, such as S-SMOTE, MDO, and SMOM, generally achieve lower mGM scores than the latter methods, indicating both improvement and continuity in the research. To this end, we can observe that local-based oversampling is effective in addressing the multiclass imbalance problem, particularly in handling small disjuncts within the dataset. It even performs better when combined with a selective instance mechanism rather than oversampling on randomly selected samples. Additionally, removing overlapped instances from majority classes is also necessary to reduce the impact of class overlap. Therefore, better methods can be proposed in the future based on the evaluation of the current state-of-the-art techniques in multiclass imbalance learning.

8.1.2 Comparison with baseline performance

Almost all data resampling methods in previous works do not provide a comparison with the baseline performance, which is crucial to ensure that the resampling method does not produce lower performance than the baseline. Table 11 compares each resampling method with the baseline performance, as measured by the mGM metric. Experiments have been conducted to obtain baseline scores by training and testing each dataset using the C4.5 algorithm through 10-fold cross-validation, repeated five times to get comprehensive results.

Table 11 indicates that MC-RBO, MC-CCR, NROMM, and MC-NRO successfully improve the performance on datasets with gradual imbalance and multi-majority types.

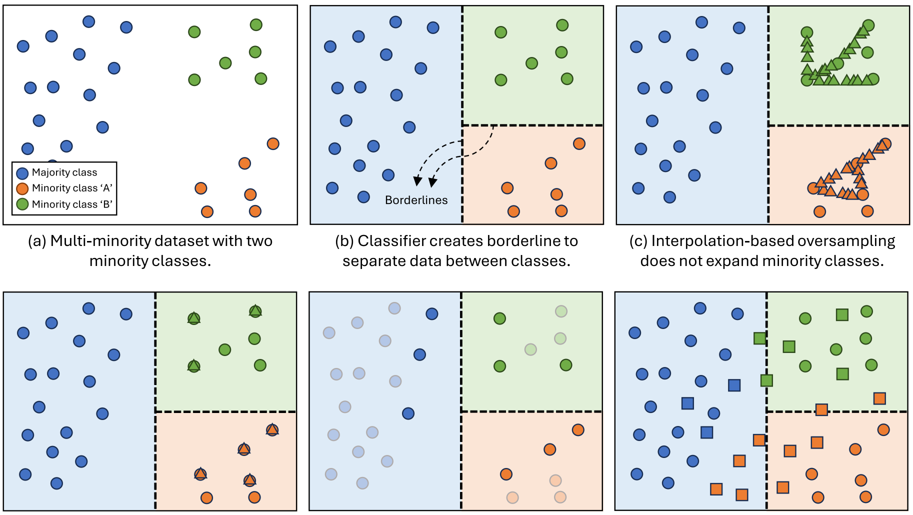
Table 11 Comparison of data resampling methods to the baseline performance based on the mGM metric. Positive improvements are written in black, while negative ones are in **bold**

Dataset	Baseline	S-SMOTE	MDO	SMOM	SOUP	MC-RBO	MC-CCR	OC-SV-US	NROMM	MC-NRO
Gradual imbalance										
Winequality-red	0.382	**-0.015**	+0.025	+0.044	+0.089	+0.101	+0.085	+0.029	+0.097	+0.107
Yeast	0.450	+0.074	+0.061	+0.079	+0.001	+0.131	+0.119	NA	+0.109	+0.141
Glass	0.692	**-0.066**	**-0.012**	**-0.012**	**-0.025**	+0.039	+0.019	NA	+0.105	+0.115
Ecoli	0.689	**-0.014**	+0.021	**-0.054**	+0.046	+0.038	+0.021	NA	+0.028	+0.049
Dermatology	0.922	+0.030	**-0.001**	**-0.001**	+0.040	+0.035	+0.027	**-0.009**	+0.042	+0.045
Flare	0.609	+0.010	**-0.023**	**-0.005**	**-0.043**	+0.046	+0.038	+0.015	+0.031	+0.054
Multi-majority										
Contraceptive	0.462	**-0.028**	+0.055	+0.064	+0.073	+0.048	+0.035	+0.019	+0.066	+0.067
Balance-scale	0.557	**-0.033**	+0.002	+0.030	+0.028	+0.062	+0.065	+0.310	+0.110	+0.087
Led7digit	0.697	+0.011	+0.085	+0.076	+0.081	+0.053	+0.107	+0.010	+0.089	+0.100
Lymphography	0.705	**-0.122**	+0.026	+0.035	NA	+0.086	+0.069	NA	+0.177	+0.138
Hayes-Roth	0.856	**-0.010**	**-0.024**	+0.012	**-0.021**	**-0.037**	**-0.004**	**-0.067**	+0.056	+0.025
Multi-minority										
Cleveland	0.336	**-0.090**	**-0.071**	**-0.090**	**-0.033**	+0.004	**-0.026**	**-0.016**	+0.024	+0.009
Newthyroid	0.932	**-0.067**	**-0.006**	+0.003	**-0.024**	+0.035	**-0.010**	**-0.028**	+0.045	+0.048
Page Blocks	0.833	**-0.089**	**-0.042**	**-0.061**	+0.051	+0.040	**-0.007**	**-0.049**	+0.049	+0.051
Wine	0.936	**-0.002**	**-0.035**	**-0.045**	**-0.026**	**-0.019**	**-0.007**	+0.048	+0.035	+0.015
Thyroid	0.980	**-0.161**	**-0.221**	**-0.203**	**-0.058**	**-0.111**	**-0.165**	+0.001	**-0.019**	**-0.049**
Zoo	0.983	**-0.348**	**-0.225**	**-0.206**	NA	**-0.133**	**-0.140**	NA	**-0.014**	**-0.046**

SOUP also achieves good performance on both types of datasets. SMOM perfectly improves the performance of all datasets in the multi-majority type, although it fails to improve most datasets in gradual imbalance. Its performance is slightly better than MDO. MC-NRO and NROMM produce positive improvement in 88% of all datasets, while MC-RBO and MC-CCR improve the performance in 76% and 59% of all datasets, respectively.

However, all resampling methods fail to improve datasets in the multi-minority type, except MC-NRO and NROMM. Earlier methods, such as S-SMOTE, MDO, and SMOM, are the poorest performers and exhibit very high negative performance in multi-minority datasets, suggesting that handling multi-minority datasets through data resampling is more challenging than handling multi-majority ones. The primary reason for this issue is the inability of resampling methods to expand the area of minority classes, as these classes lack sufficient representative instances for classifiers to learn from, resulting in a low generalization ability. This issue is illustrated in Fig. 12. Datasets with multi-minority configurations tend to produce poor borderlines between classes, resulting in the misclassification of unseen instances (see Fig. 12f).

Interpolation-based oversampling methods, such as S-SMOTE, SMOM, and SCUT, tend to fail in expanding the minority class area because they only generate synthetic instances between two linearly adjacent instances (see Fig. 12c). This is even more problematic for duplication-based oversampling methods, such as SOUP, or data filtering methods, like OCSV-US, as they lack expansion capability, as illustrated in Figs. 12d and 12e, respectively. Wang and Yao (2012) asserted that when a dataset contains many minority classes, expanding each area of the minority class is more important than reducing the area of the majority class through undersampling. Hence, expanding the area of minority classes may increase the classifier's probability of learning the untouched areas within these classes and help the classifier better classify unseen instances (test data) in the future.



(a) Multi-minority dataset with two minority classes.
(b) Classifier creates borderline to separate data between classes.
(c) Interpolation-based oversampling does not expand minority classes.
(d) Duplication-based oversampling does not expand minority classes
(e) Data filtering does not expand minority classes
(f) Poor borderlines misclassify data from minority classes

Fig. 12 Illustration of poor borderlines that result in misclassification to unseen instances (test data) in minority classes. Triangles represent the synthetic instances, and rectangles represent the unseen instances

The reason above explains why most data resampling methods fail in multi-minority datasets. MC-CCR can expand the area of minority classes. However, further analysis reveals that the data translation mechanism in MC-CCR may disarrange data distribution in minority classes, as the translation occurs in all classes except the smallest one. When two consecutive minority classes are paired for resampling, the area of the larger minority class may be reduced by the smaller one, thereby enlarging the area of one minority class while simultaneously decreasing the area of the other minority classes. OCSV-US, which utilizes only undersampling, performs slightly better than MC-CCR on the Wine and Thyroid datasets. Even though it does not enlarge the area of minority classes, it does not shrink their areas.

In contrast, MC-NRO and NROMM achieve the best performance on multi-minority datasets. Both methods successfully improve four out of six multi-minority datasets. MC-RBO with appropriate parameters also improves three multi-minority datasets. This compelling performance is achieved due to the algorithms' ability to enlarge the area of the minority class without shrinking the areas of other classes.

Overall, Fig. 13 indicates that MC-NRO and NROMM are the best data resampling methods. Both have the highest improvement with the lowest degradation in performance. This result implies that resampling methods with local-based and boundary-based oversampling mechanisms generally perform better than those without these mechanisms. Besides, using SMOTE-based oversampling with several constraints, as in NROMM, can also be effective. Moreover, the performance improves when certain undersampling with data removal procedures are applied to reduce the negative impact of class overlap, as in MC-NRO and NROMM. In contrast, pseudo undersampling, such as data translation in MC-CCR, should not be applied to all classes, as it can result in substantial performance degradation, as shown in Fig. 13.

8.1.3 Evaluation of data resampling with respect to difficulty factors

Further analysis compares the latter method in Table 10, i.e., SOUP, MC-RBO, MC-CCR, OCSV-US, NROMM, and MC-NRO. The comparison is based on the difficulty factors: imbalance ratio, percentage of class overlap, number of classes, and class configuration. As depicted in Fig. 14, resampling methods applied on datasets with a low imbalance ratio, a low class overlap rate, or a small number of classes constantly achieve higher performance. The performance tends to decrease gradually as the value of those three aspects increases.

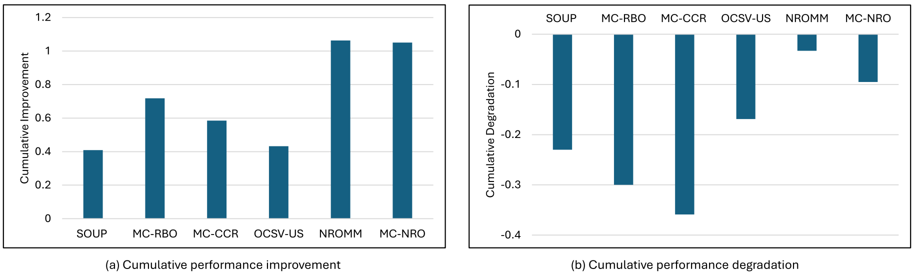


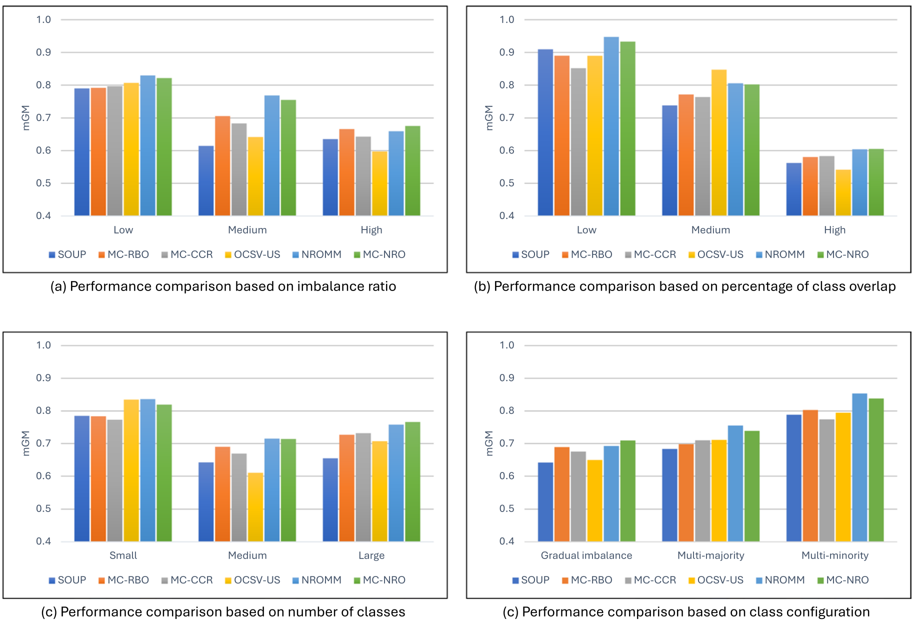
Fig. 13 Comparison of cumulative performance **a** improvement and **b** degradation of several data resampling methods on different types of multiclass imbalanced datasets

Consistently, MC-NRO and NROMM tend to outperform other methods across datasets with varying imbalance ratios, overlap rates, and numbers of classes.

Among the three aspects, the percentage of class overlap is the most significant factor that deteriorates classification performance, followed by the class configuration factor. Figure 14 also shows that classification performance on multi-minority datasets tends to be higher than on other class configurations. This happens because multi-minority datasets have a small percentage of class overlap. Nevertheless, most data resampling methods failed to further improve their classification performance. As the percentage of overlap increases, as in gradual imbalance and multi-majority, classification performance also tends to decline.

8.2 Effectiveness of ensemble learning approach

Research continuity in ensemble learning is relatively poor compared to data resampling. All methods in Tables 5 and 6 do not provide any performance comparison with each other. Each method only compares its performance with classical ensemble algorithms that are mostly not dedicated to multiclass imbalance learning, such as RUSBagging (Wang and Yao 2009), SMOTEBagging (Wang and Yao 2009), RUSBoost (Seiffert et al. 2010), EasyEnsemble (Liu et al. 2006), and AdaBoost.NC (Wang et al. 2010). Therefore, this paper attempts to collect ensemble algorithms developed for multiclass imbalance learning and compare their performance with one another. However, this comparison may not be as accurate as previous resampling methods; nevertheless, it may provide useful information for ongoing work in this field. The comparison is divided into two parts: comparing the methods and evaluating each method's performance against a baseline.



(a) Performance comparison based on imbalance ratio

(b) Performance comparison based on percentage of class overlap

(c) Performance comparison based on number of classes

(c) Performance comparison based on class configuration

Fig. 14 Comparison between the latter resampling methods: SOUP, MC-RBO, MC-CCR, OCSV-US, NROMM, and MC-NRO according to **a** imbalance ratio, **b** percentage of class overlap, **c** number of classes, and **d** class configuration

8.2.1 Comparison between methods

Based on Table 12, of the 12 ensemble algorithms reviewed in this work only six can be compared fairly with respect to the evaluation method, performance metric, and classifier used. The comparison is based on the AvAcc metric, which has been provided in most previous works. Surprisingly, Easy-BPNN, which uses the OvO decomposition strategy, outperforms all newer methods, achieving the highest performance on 33% of all datasets–most of which are multi-minority datasets. Easy-BPNN also performs well on multi-majority datasets and several highly overlapping datasets, such as Winequality-red, Contraceptive, and Cleveland.

The combination of the bagging mechanism and the AdaBoost algorithm, integrated with backpropagation neural networks (BPNN), yields significant performance improvements for Easy-BPNN. Compared to the decision tree algorithm commonly used in other ensemble frameworks, BPNN tends to be superior, as it has a better non-linear nature. Moreover, the undersampling mechanism applied in each bootstrap helps reduce the impact of class overlap while increasing the diversity of the classifier. However, the computational cost of Easy-BPNN may increase significantly due to the training phase of AdaBoost and BPNN.

On the other hand, DRCW-ASEG and DES-MI show competitive performance with each other. DRCW-ASEG is slightly better on gradual imbalance and multi-minority datasets, while DES-MI shows better performance on multi-majority datasets. DRCW-ASEG has an adaptive oversampling mechanism that is applied only during the testing phase for each unseen instance. This helps the method balance the number of instances across all classes around the queried instance, thus improving the final prediction results.

Table 12 Comparison of ensemble learning performance based on the AvAcc metric. A darker color indicates a higher score

Dataset	Easy-BPNN	DRCW-ASEG	DES-MI	MultiRandBal	DPSE	DOMIN
Gradual imbalance						
Winequality-red	0.420	NA	**0.429**	0.390	0.394	NA
Yeast	NA	**0.588**	0.583	0.584	0.541	0.450
Glass	0.700	**0.783**	0.765	0.709	0.712	0.588
Ecoli	NA	0.767	0.727	0.615	0.595	**0.785**
Dermatology	0.968	**0.970**	0.960	0.967	0.942	0.940
Flare	0.651	0.609	**0.652**	0.618	0.623	NA
Multi-majority						
Contraceptive	**0.520**	0.488	0.506	0.514	0.487	0.492
Balance-scale	**0.883**	0.578	0.639	0.613	0.581	0.725
Led7digit	0.720	0.696	**0.733**	0.713	0.705	NA
Lymphography	0.862	**0.884**	0.847	0.775	0.830	0.729
Hayes-Roth	0.787	0.877	**0.882**	0.855	0.863	0.708
Multi-minority						
Cleveland	0.312	0.302	0.306	0.310	**0.316**	NA
Newthyroid	**0.979**	0.949	0.939	0.936	0.940	0.281
Page Blocks	**0.921**	0.817	0.778	0.749	0.890	0.688
Wine	**0.986**	0.972	0.962	0.934	0.942	0.910
Thyroid	**0.968**	NA	0.919	0.961	0.951	0.233
Zoo	0.894	**0.925**	0.918	0.838	0.924	NA
Shuttle	0.879	NA	0.918	0.871	**0.959**	0.620

In DRCW-ASEG, synthetic samples are generated only when the classes in the neighborhood of the queried instance have a significantly high imbalance ratio. This strategy, however, helps DRCW-ASEG avoid overfitting caused by excessive oversampling. On the other hand, DES-MI is prone to over-oversampling using SMOTE, especially on datasets with numerous minority classes, such as those with gradual imbalance and multi-minority datasets. This explains why DRCW-ASEG performs better on gradual imbalance and multi-minority datasets. However, DRCW-ASEG does not achieve optimal performance on certain highly overlapping datasets, such as Cleveland and Contraceptive. Moreover, it requires many user-defined parameters that must be carefully tuned.

DES-MI shines in its performance on specific datasets, such as Winequality-red and Hayes-Roth, which exhibit a high percentage of class overlap. In contrast, MultiRandBal's overall performance is notably lower than DES-MI despite both algorithms utilizing a random balance mechanism in the sampling process. This suggests that the random balance version used in DES-MI, which randomizes the size of each subset, is more effective than the one used in MultiRandBal, which forms each subset with equal size. Additionally, each subset in DES-MI maintains a balanced class distribution, while MultiRandBal allows each subset to remain imbalanced. As shown in Table 12, MultiRandBal's performance is considered moderate among all methods.

In contrast to DES-MI, DPSE has lower performance on gradual imbalance and multi-majority datasets but significantly better on multi-minority datasets. The bootstrapping process in DPSE also generates subsets of different sizes, just as DES-MI. However, the OvO mechanism adopted by DPSE may contribute to the deterioration of classification performance on gradual imbalance and multi-majority datasets. Meanwhile, DOMIN only achieves the highest performance on the Ecoli dataset and has the lowest performance among all methods for most datasets.

To sum up, EasyEnsemble, which combines bagging and boosting mechanisms, achieves optimal performance, as confirmed by the results of Easy-BPNN presented in Table 12. On the other hand, the resampling strategy employed during the testing phase, as in DRCW-ASEG, is also promising for achieving good classification results, although the high computational cost associated with the testing phase must be taken into consideration. Meanwhile, the bagging mechanism, which uses different sizes of subsets with balanced class distribution, as in DES-MI and DPSE, also achieves good performance on certain types of datasets. This strategy helps increase the diversity of base classifiers, thus improving the generalization ability of the classification model. In future work, several strategies from the methods mentioned above can be combined to develop a better ensemble framework.

8.2.2 Comparison with baseline performance

Table 13 signifies that the ensemble learning methods generally achieve higher scores than the baseline performances on almost all datasets. Easy-BPNN successfully combines EasyEnsemble with backpropagation neural networks as the base classifier, significantly improving the classification performance by an average of 7.6%. Easy-BPNN makes the most significant improvement on the Balance-scale dataset with a 32% higher score, the highest among all methods. It even produces the highest improvement among others on multi-majority datasets. However, this algorithm may take a long time to train since it is an ensemble of AdaBoosts with neural networks as the base classifier.

On the other hand, DRCW-ASEG and DES-MI have similar improvements, with an average of 6.3% higher than the baseline. MultiRandBal and DPSE also exhibit comparable performance improvements on gradual imbalance and multi-majority datasets, while DPSE is significantly better on multi-minority datasets. DPSE performs similarly to Easy-BPNN on multi-minority datasets. The dynamic partition ensemble mechanism in DPSE effectively strengthens the representation of minority classes.

Similar to data resampling, the performance of ensemble learning on multi-minority datasets is less effective than on gradual imbalance and multi-majority datasets. Some multi-minority datasets are difficult to handle. All ensemble learning methods failed to improve the baseline performance on Thyroid and Zoo, as both datasets already have very high AvAcc, at 98.30% and 99.30%, respectively. These two datasets do not require imbalance handling to obtain an optimal classification model. Additionally, they have a relatively small percentage of class overlap, making it easier for the classifier to distinguish between the classes.

Further analysis reveals that DES-MI exhibits comparable performance to Easy-BPNN in enhancing the baseline performance. Based on Fig. 15, DES-MI and Easy-BPNN achieve the highest cumulative improvement with the least degradation in improvement among other ensemble learning methods. Both methods exhibit high diversity in their base classifiers, which enhances the generalization capability of ensemble learning. This diversity can be achieved by generating subsets of different sizes, utilizing another ensemble algorithm within the proposed ensemble learning framework, and employing specific data preprocessing strategies.

Table 13 Comparison of ensemble learning to the baseline performance based on the AvAcc metric. Positive improvements are written in black, while negative ones are in **bold**

Dataset	Baseline	Easy-BPNN	DRCW-ASEG	DES-MI	MultiRandBal	DPSE	DOMIN
Gradual imbalance							
Winequality-red	0.362	+0.058	NA	+0.067	+0.028	+0.032	NA
Yeast	0.445	NA	+0.143	+0.138	+0.139	+0.096	+0.005
Glass	0.680	+0.020	+0.103	+0.085	+0.029	+0.032	**-0.092**
Ecoli	0.629	NA	+0.138	+0.098	**-0.014**	**-0.034**	+0.156
Dermatology	0.927	+0.041	+0.043	+0.033	+0.040	+0.015	+0.013
Flare	0.599	+0.052	+0.010	+0.053	+0.019	+0.024	NA
Multi-majority							
Contraceptive	0.459	+0.061	+0.029	+0.047	+0.055	+0.028	+0.033
Balance-scale	0.565	+0.318	+0.013	+0.074	+0.048	+0.016	+0.160
Led7digit	0.686	+0.034	+0.010	+0.047	+0.027	+0.019	NA
Lymphography	0.671	+0.191	+0.213	+0.176	+0.104	+0.159	+0.058
Hayes-Roth	0.843	**-0.056**	+0.034	+0.039	+0.012	+0.020	**-0.135**
Multi-minority							
Cleveland	0.294	+0.018	+0.008	+0.012	+0.016	+0.022	NA
Newthyroid	0.925	+0.054	+0.024	+0.014	+0.011	+0.015	**-0.644**
Page Blocks	0.833	+0.088	**-0.016**	**-0.055**	**-0.084**	+0.057	**-0.145**
Wine	0.928	+0.058	+0.044	+0.034	+0.006	+0.014	**-0.018**
Thyroid	0.983	**-0.015**	NA	**-0.064**	**-0.022**	**-0.032**	**-0.750**
Zoo	0.993	**-0.099**	**-0.068**	**-0.075**	**-0.155**	**-0.069**	NA
Shuttle	0.877	+0.002	NA	+0.041	**-0.006**	+0.082	**-0.257**

8.2.3 Evaluation of ensemble learning with respect to difficulty factors

Figure 16 shows that the performance of ensemble learning is quite similar to that of data resampling methods. Ensemble learning is slightly better than data resampling, as it maintains relatively constant performance across any imbalance ratio and number of classes. The diversity of classifiers within bagging and boosting mechanisms is the main factor behind this better performance. Among the ensemble learning methods, Easy-BPNN consistently demonstrates the best performance across all variations of difficulty factors.

However, it is worth noting that the percentage of class overlap still plays the most significant role in deteriorating classification performance. As the overlap percentage increases, the classification performance tends to decrease. The use of SMOTE as the data preprocessing strategy in most ensemble learning methods is responsible. Therefore, future work can address this issue by integrating more effective data resampling methods into the ensemble learning mechanism. Furthermore, multi-minority datasets demonstrate higher classification performance due to their low class overlap. Despite this, most ensemble learning frameworks have failed to deliver additional performance gains on these datasets.

9 Lessons learned and future work

Important points extracted from the discussions in the previous sections constitute valuable lessons for future works:

1. Class overlap and class configuration are the most significant difficulty factors impacting the classification performance of multiclass imbalanced datasets. The overlap between majority classes (MajMaj) and between majority and minority classes (MajMin) has a significant impact on the deterioration of classification performance. Meanwhile, datasets with a multi-minority configuration are more difficult to handle using data resampling than those with a multi-majority or gradual imbalance. Almost all data resampling methods cannot improve performance on most multi-minority datasets.
2. Oversampling using synthetic data generation based on the local boundary around observed instances is an effective approach for handling multiclass imbalance problems. This strategy is successfully applied in MC-RBO, OREM-M, and MC-NRO.

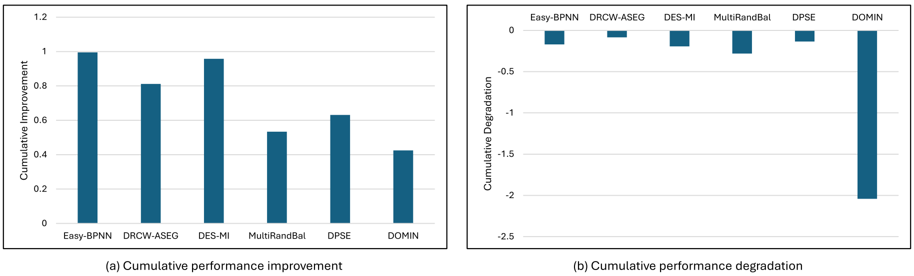
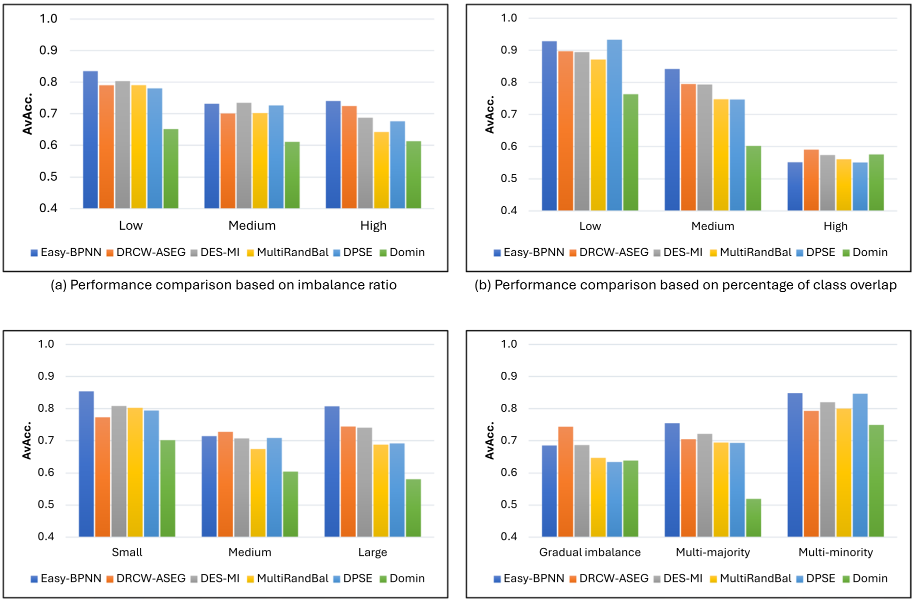


Fig. 15 Comparison of cumulative performance **a** improvement and **b** degradation of several ensemble learning methods on different types of multiclass imbalanced datasets



(a) Performance comparison based on imbalance ratio

(b) Performance comparison based on percentage of class overlap

(c) Performance comparison based on number of classes

(c) Performance comparison based on class configuration

Fig. 16 Comparison between several ensemble learning methods according to **a** imbalance ratio, **b** percentage of class overlap, **c** number of classes, and **d** class configuration

3. Performing data resampling with class-pairwise handling is more effective than using per-class or all-classes handling. This is demonstrated by MC-RBO, MC-CCR, NROMM, and MC-NRO, all of which employ pairwise handling.
4. Categorizing instances into safe, borderline, and overlap (or inland, borderline, and trapped) is quite effective in determining the priority order in the oversampling and undersampling processes. This mechanism may improve performance on several resampling techniques, such as NROMM and SOUP.
5. The undersampling technique is effective in handling the high class overlap in gradual imbalance and multi-majority datasets. Meanwhile, oversampling by generating synthetic instances based on local boundaries is effective in multi-minority datasets since it can enlarge the area of the minority class.
6. Data translation in MC-CCR is not optimal for multi-minority datasets because it is applied to all classes, which makes it highly likely that instances from minority classes will also be shifted. Moreover, data translation may introduce another issue by causing class overlap as a result of the data movement. The first problem can be addressed by restricting data translation to majority classes only, while the second problem can be mitigated by employing data removal to reduce the impact of class overlap, as proposed in MC-NRO.
7. The ensemble learning method based on the boosting mechanism has proven highly effective in achieving optimal classification performance. Furthermore, as in DES-MI, bagging mechanisms with varying subset sizes have shown superior performance compared to using a uniform subset size. The higher diversity of the base classifiers in

ensemble learning further enhances its generalization capability, instilling confidence in the robustness of these techniques.

Multiclass imbalance learning, however, is still less explored than binary class imbalance. Difficulty factors, such as multiclass overlap, small disjunct, and class configuration, require an appropriate strategy to improve the classification performance. Based on the review and analysis from the previous methods of data resampling and ensemble learning, the following are several research gaps that can be explored to enhance the performance and robustness of classification models, particularly in the context of multiclass imbalance learning:

1. Data resampling and ensemble learning have shown potential to improve performance on multi-minority datasets, but their effectiveness has not yet been fully optimized. This highlights the pressing need for more effective strategies to improve classification performance, especially on multi-minority datasets.
2. Handling multiclass imbalance at the data level is primarily dominated by data resampling mechanisms. However, feature selection and feature extraction, which have been scarcely discussed in previous studies, present an opportunity for diversification and exploration. Therefore, an in-depth exploration of feature selection and feature extraction is needed to determine their impact on classification performance in multiclass imbalance learning.
3. Several oversampling and undersampling techniques from various studies have proved effective in handling multiclass imbalanced datasets. Combining these methods to examine their impact on classification performance presents an opportunity for improvement.
4. Almost all ensemble learning methods use SMOTE as the data processing method to balance class distribution. Since SMOTE has several limitations, further research is needed using more effective data resampling methods to enhance the performance of ensemble learning.
5. Most ensemble learning methods employ the majority voting mechanism to produce the final prediction. Weighted and soft voting mechanisms can be explored to determine their impact on classification performance in multiclass imbalance learning.
6. Decision trees are the most widely used base classifiers in ensemble learning frameworks due to their efficiency and non-linearity. However, the use of other machine learning algorithms capable of addressing class imbalance issues, such as RBFNN, can be explored in future work.

10 Conclusion

This paper reviewed various data resampling and ensemble learning methods in the context of multiclass imbalance learning. It discussed 14 data resampling methods and 12 ensemble learning techniques. Comparative analyses were conducted to evaluate the effectiveness of each algorithm in improving classification performance on various datasets.

We also provided an in-depth review of ad hoc methods in data resampling and ensemble learning to address the challenges of multiclass imbalance learning. Each method was detailed to provide insights into its mechanisms, advantages, and limitations. Additionally, we reviewed 19 commonly used datasets involving small experiments to measure baseline performance. These datasets are categorized into three groups based on class distribution: gradual imbalance, multi-majority, and multi-minority. We then presented a comparative analysis of several data resampling and ensemble learning methods against various difficulty factors in multiclass imbalance learning. This analysis highlights the strengths and weaknesses of each method, revealing important lessons and research gaps for future exploration.

However, this study has limitations. The comparative performance analysis relied on results from previous studies, including the evaluation metrics and datasets used. The paper did not address the relationship between classification performance and the number of features in the dataset. This paper also did not provide an analysis of small disjuncts in each dataset and their impact on the classification performance of each method. A more detailed investigation is needed to understand how the number of features and the presence of small disjunct affect the performance of different methods.

Moreover, we were unable to compare all the methods discussed in this article due to the limitations of the experimental results provided in the previous works. This paper also focuses only on static datasets and excludes the review of imbalance methods in data streams. Based on our observation, this issue has its own challenges and approaches that differ from those in static data and is, therefore, more appropriately treated as a separate research track in the field.

In summary, this paper contributes to understanding the workings and effectiveness of various data resampling and ensemble learning methods in addressing multiclass imbalance problems. Overall, advancements in these methods show positive improvements in multiclass imbalance learning performance. Future research should aim to achieve better performance, particularly on multi-minority datasets and those with high class overlap.

Acknowledgements Muhammad Fachrie thanks the Center for Higher Education Funding and Assessment (PPAT), the Ministry of Higher Education, Science, and Technology of the Republic of Indonesia, and the Indonesia Endowment Fund for Education (LPDP) for funding his doctoral study through the Indonesian Education Scholarship (BPI) with grant no. 02166/J5.2.3./BPI.06/9/2022.

Author contributions M. F.: Conceptualization, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing - original draft, Writing - review & editing. A. M.: Formal analysis, Methodology, Supervision, Validation, Writing - review & editing. R. P.: Conceptualization, Formal analysis, Methodology, Supervision, Validation, Writing - review & editing.

Funding No funding to report.

Data availability No datasets were generated or analysed during the current study.

Code availability Not applicable.

Materials availability Not applicable.

Declarations

Conflict of interest The authors declare that there are no Conflict of interest relevant to this article.

Ethical approval and consent to participate Not applicable.

Consent for publication Not applicable.

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